MCAP: Deployment-Time Layer Profiling for Memory-Constrained LLM Inference

Anurita Das
Genovation Technological Solutions Pvt Ltd
anurita@genovationsolutions.com

Abstract

Deploying large language models to heterogeneous hardware is constrained by memory, not compute: a 4-bit 8B model still requires ∼5–6 GB of resident VRAM, ruling out most consumer GPUs, mobile accelerators, and older cloud silicon. Existing post-training quantization fixes the precision choice at calibration time, so the set of devices on which a given quantized model is viable is decided before it ever reaches hardware. We introduce a *load-time control layer* for LLM inference: the per-layer precision decision is made on the target device from a lightweight runtime signal rather than fixed offline, which substantially widens the set of reachable operating points for a given set of weights.

We introduce **MCAP** (Monte Carlo Activation Profiling), a load-time layer importance estimator: a 60-second gradient-free profiler that produces a per-layer importance signal from 12 calibration prompts, and **NVE**, a Rust+CUDA inference engine that uses that signal to drive two coupled per-layer decisions: precision dispatch (W4A8 vs. W4A16) and residency tier (GPU / RAM / SSD). A profile is a 338-byte JSON keyed by architecture rather than a new set of weights, and each target computes its own on load. This couples two decisions that prior systems handle separately: precision routing and memory placement, from a single runtime signal.

NVE delivers 1.5–1.8× higher decode throughput than llama.cpp Q4_0 on NVIDIA T4 across 1B, 3B, and 8B Llama models, with the advantage over W4A16 widening from 2.3× at 1B to 2.9× at 8B. More importantly, the same signal drives a three-tier weight pager that reaches operating points conventional GPU-resident inference cannot: Llama-3.2-3B (∼6.4 GB BF16) runs in 2 GB at 2.31 tok/s, and Llama-3.1-8B (∼16 GB BF16) runs in 4 GB at 1.25 tok/s, both with no observable degradation on the evaluated tasks relative to the unconstrained baseline. Against AutoAWQ at matched 4-bit precision, MCAP is within binomial CI on HellaSwag (54% vs. 52%) with no offline calibration and no weight modification. A single 338-byte profile, SHA-256-identical across six parallel containers on two NVIDIA silicon generations (Turing T4 and Ampere A10G), drives correct 87.5% task accuracy on all 12 runs at 0 ms on-device profiling cost. We report where the method breaks (aggressive profile-guided quantization fails at 1B scale) and treat cross-silicon validation on Jetson, Apple Silicon, and consumer RTX as the natural follow-up, enabling LLM deployment across previously infeasible memory regimes without modifying model weights.

1 Introduction

The operational question in LLM inference today is not only how much a model can be compressed but which devices can run it at all. A 4-bit 8B model still requires ∼5–6 GB of resident VRAM; a 3B model in BF16 requires ∼7 GB. These footprints place most consumer GPUs, mobile accelerators, integrated graphics, and older cloud silicon outside the feasible region. Existing post-training

quantization commits the precision choice at calibration time and emits a new set of weights keyed to that choice, so the set of viable targets for a given model is fixed before deployment. In a setting where target hardware is heterogeneous and individual device budgets vary across sessions (thermal throttling, battery state, other processes), this binding is the limiting factor.

We move the per-layer precision decision from calibration time to load time. A short forward pass over a handful of calibration prompts is enough to rank layers by their sensitivity to precision reduction; the ranking is computed on the target device in seconds, cached by architecture, and reused across every subsequent load. From that single signal we drive two per-layer decisions: precision dispatch (W4A8 vs. W4A16) and residency tier (GPU / CPU RAM / SSD). The same weights serve multiple memory budgets, and the set of reachable operating points widens accordingly.

Contributions.

1. **MCAP, a load-time per-layer importance signal** (Section 3.1). A 60-second, gradient-free, weight-free profiler over 12 calibration prompts (floor of 8 for stable top-k recovery, tunable count and domain, extensible to live traffic). Scores are hardware-independent and cached per architecture. Across eight architectures from GPT-2 (0.1B) through Qwen2-7.6B, MCAP recovers a strongly non-uniform layer-importance pattern that a static "protect the last layer" rule does not reproduce.

2. **Importance-guided execution coupling precision and residency** (Section 3.2, Section 3.3). At decode time each layer is routed on its MCAP score to a precision tier (W4A8 or W4A16) and a residency tier (GPU, CPU RAM, or SSD), driven by one signal rather than two independent systems. The W4A8 path delivers 2.3–2.9× over W4A16 and 1.5–1.8× over llama.cpp Q4_0 on T4, with the gap widening at larger scale (Table 7). The three-tier pager runs models 3.5–4× larger than GPU memory (Llama-3.2-3B in 2 GB, Llama-3.1-8B in 4 GB) with no observable degradation on evaluated tasks relative to the unconstrained baseline, and raises throughput by 9–58% over the unconstrained run in several configurations.

3. **An integrated Rust+CUDA implementation spanning 12+ architectures** (Section 3). 22K lines of Rust and 1,413 lines of hand-fused CUDA (RMSNorm+RoPE, fused QKV projections, flash decode attention, W4A8, W4A16) behind a unified `GenericBlockWeights` abstraction covering Llama, Qwen, Phi, Gemma, GPT-2, Falcon, and GPT-NeoX. MCAP profiles and per-layer dispatch work unchanged across all supported architectures.

Why the three components are coupled. MCAP provides the signal that makes adaptive execution possible; routing and paging are what turn that signal into throughput and feasibility. Routing alone reduces to a static rule that mis-fires on architectures where the rule is wrong, and paging alone is LRU without an importance prior. The strongest results appear where the three reinforce each other, in the regime they were designed for: memory budgets at which conventional inference is infeasible.

Feasibility first, throughput second. The primary result is about which operating points are reachable. NVE reaches points that GPU-resident inference cannot at any precision: Llama-3.2-3B with full BF16 weights ($\sim$7 GB) runs in 2 GB with no observable degradation on the evaluated task suite, and Llama-3.1-8B ($\sim$16 GB BF16) runs in 4 GB while matching its constrained baseline on the same suite. Four-bit quantization alone still requires $\sim$5–6 GB for an 8B model, so the 4 GB point is a new hardware class rather than an incremental improvement. Throughput is the secondary result: at matched arithmetic precision the W4A8 routing path delivers 1.5–1.8$\times$ over llama.cpp Q4_0, and W4A16, W4A8, and MCAP Mixed show no measurable difference on WikiText-2 and HellaSwag at the evaluated sample sizes.

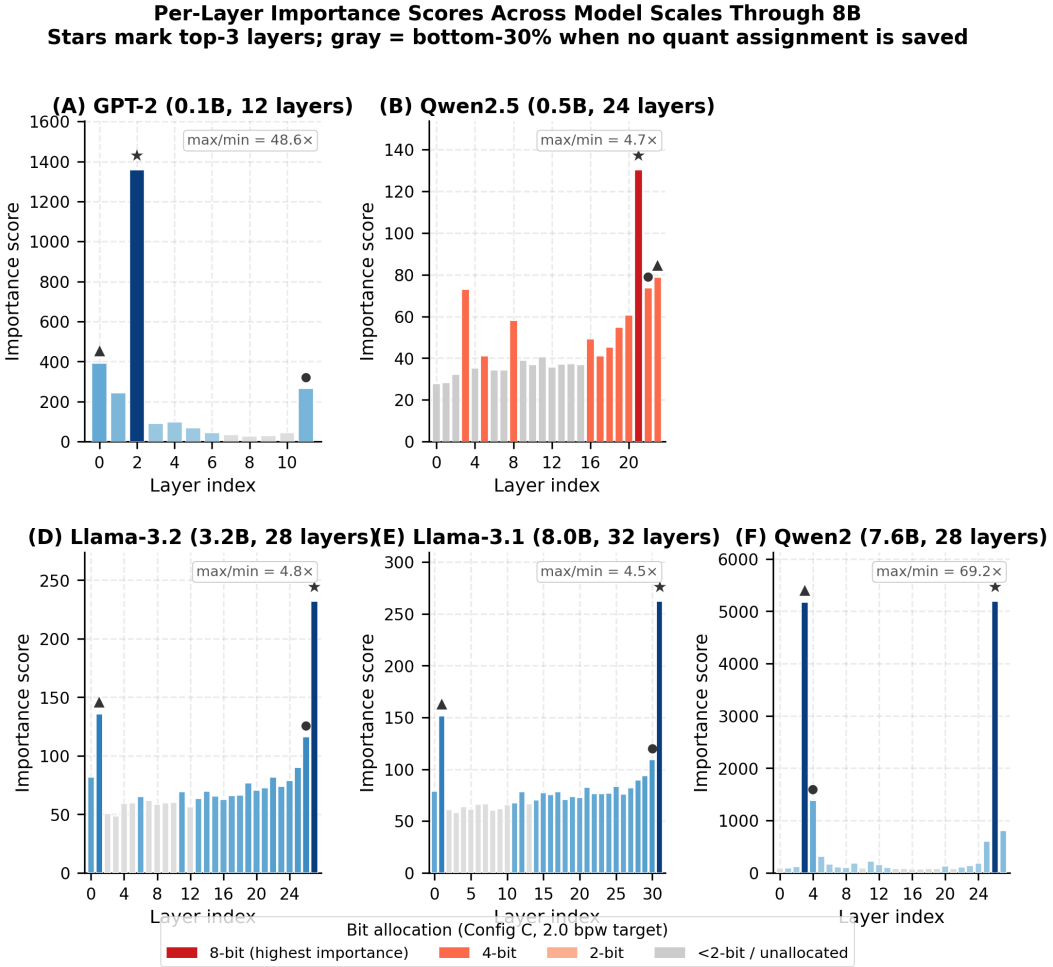

Figure 1: **MCAP importance scores across model scales through 8B.** Per-layer importance scores for GPT-2 (0.1B), Qwen2.5 (0.5B), Llama-3.2-1B, Llama-3.2-3B, Llama-3.1-8B, and Qwen2-7.6B. The dominant outlier layer is usually near the end of the network, but the exact pattern is architecture-dependent: most models show a single final-layer outlier, while larger Qwen variants exhibit additional outliers.

Figure 1 makes the profiling claim concrete. The non-uniformity is stable from 0.1B through 8B,

but the *pattern* is not rigid: most Llama-family runs show one dominant late-layer peak; GPT-2 and larger Qwen variants show several. This is the reason a runtime signal is preferable to a static rule such as "always protect the last layer": the rule is correct on average but wrong on the architectures where it is wrong.

Quantitative consequences. On NVIDIA T4, the throughput gap corresponds to a 44% reduction in tok/s^{-1} cost vs. llama.cpp at 1B (Section 4.2). The W4A8 advantage over W4A16 grows from 2.3× at 1B to 2.9× at 8B, consistent with a bandwidth-limited regime rather than a small-model effect. The memory side reaches points not accessible to GPU-resident inference. The same MCAP signal drives three execution modes on a single spectrum (Section 3.5): *hot-only* (skip low-importance layers entirely; no paging traffic), *hot-only+AWQ* (saliency-weighted quantization on the retained layers for higher quality when the budget fits), and *paging* (GPU→RAM→SSD) for budgets below the hot-only viability floor, reaching full-BF16 3B in 2 GB and 8B in 4 GB with no observable degradation on the evaluated task suite (Section 4.6).

2 Background and Related Work

This paper focuses on the load-time decision problem: how per-layer precision and memory placement should be chosen for a given set of weights under varying hardware constraints. We survey two threads that bear directly on that problem: post-training quantization (which decides precision at calibration time) and weight offloading / memory-efficient inference (which decides placement, usually at inference time). The contribution of this paper is the coupling of these two decisions on a single load-time signal.

2.1 Post-Training Quantization for LLMs

Post-training quantization (PTQ) compresses model weights, activations, and increasingly the KV cache after training, to reduce memory and compute. The field has advanced rapidly in 2022–2024 toward full-stack quantization (weights + activations + KV cache) rather than weight-only compression, and several methods already use runtime-aware components. We summarize the lines of work most relevant to this paper.

Weight-only quantization. GPTQ Frantar et al. [2022] uses approximate second-order information (Hessian inverse) to find per-column quantization scales, achieving 3–4 bit precision with small perplexity degradation; it is explicitly designed to scale to very large models efficiently. AWQ Lin et al. [2023] observes that ∼1% of weight channels are disproportionately important and protects them via per-channel scaling; AWQ already uses activation statistics to identify salient channels and explicitly argues for cross-domain generalization without backprop or reconstruction. Its saliency signal is a form of importance estimation at finer granularity than ours. More recent methods use rotation-based outlier suppression: QuaRot Ashkboos et al. [2024] applies Hadamard rotations to flatten outliers before 4-bit quantization, and SpinQuant Liu et al. [2024] learns the rotation matrices end-to-end. HQQ Badri and Shaji [2023] uses half-quadratic splitting for calibration-free weight quantization. Marlin Frantar et al. [2024] is a reference high-throughput W4A16 GEMM kernel on Ampere/Hopper and co-designs the kernel with the weight format. These methods produce static quantized weights but often couple to online components at inference time (per-token activation handling, runtime scale application).

Activation quantization. Activation behavior, rather than weights alone, is the central bottleneck in much of the recent PTQ literature. SmoothQuant Xiao et al. [2022] argues that activations are harder to quantize than weights and migrates the difficulty between the two via a mathematically equivalent per-channel scaling that is applied online. LLM.int8() Dettmers et al. [2022] performs mixed-precision decomposition for outlier dimensions *during inference*: outlier features ($>6\sigma$) use FP16 while the remaining >99.9% use INT8. This is runtime outlier handling at the feature-dimension level, not a runtime profiling scheme at the layer level. Atom Zhao et al. [2024] co-designs low-bit serving with mixed-precision reordering, dynamic activation quantization, low-bit kernels, and KV-cache quantization for W4A4, and is the closest published W4A8-adjacent kernel to ours. Atom is stronger prior art on low-bit serving co-design; it does not perform runtime layer-level importance profiling, which is the axis we target. ZeroQuant-V2 Yao et al. [2023] reports that

activation quantization is more sensitive than weight quantization and studies layer-wise sensitivity for mixed-precision assignment; OmniQuant Shao et al. [2024] learns transformations that shift quantization difficulty between weights and activations. KIVI Liu et al. [2024] targets the KV cache, which can exceed the model weight footprint in long-context settings; it is adjacent to our work rather than directly competing with the weight-residency mechanism we use. Two features of this line of work are relevant here: activation magnitude and outlier structure are central to modern PTQ, and calibration is typically combined with online components (equivalent scaling at runtime, dynamic per-token activation quantization, runtime outlier decomposition). Our contribution is that a single load-time per-*layer* signal can drive both precision dispatch and memory residency without per-target calibration.

Weight-sensitivity methods. SqueezeLLM Kim et al. [2024] computes per-weight sensitivity from the Hessian and uses it for non-uniform quantization with a dense-sparse decomposition; SpQR Dettmers et al. [2024] isolates outlier weights offline and stores them separately. Both operate at per-weight granularity, so importance is already a well-studied quantity in PTQ; MCAP is orthogonal, operating at the coarser per-layer granularity on unmodified weights and driving paging as well as precision.

Per-layer bit allocation. The EXL2 format Turboderp [2024] used by ExLlamaV2 already implements per-layer bit allocation, chosen from offline error measurements; based on its public documentation it is the closest existing work to the per-layer granularity we use. MCAP differs in three ways: it computes the per-layer signal at load time, it selects precision per activation path (W4A8 vs. W4A16) rather than per weight, and the same scores drive paging placement. EXL2 chooses bits per weight offline; MCAP chooses per-activation precision and residency tier at load time.

KV cache and co-design. Model weights are not the only memory-dominant object in modern LLM inference. KIVI Liu et al. [2024] and related work show that the KV cache can exceed the model weight footprint in long-context settings. Several prior systems already co-design kernels or memory systems with quantization: MARLIN Frantar et al. [2024] co-designs a high-throughput W4A16 GEMM with quantization format, Atom co-designs kernels and serving behavior for W4A4, and KIVI co-designs KV-cache quantization with memory management. Our work focuses on per-layer weight-precision and weight-residency coupling, and treats KV-cache management as an orthogonal axis that can be composed with our signal (we do not quantize the KV cache in this paper; a natural extension is MCAP-guided KV precision, noted in the Conclusion).

Gap. Prior work has studied importance at weight, channel, and (in EXL2's case) layer granularity, as well as activation outliers at runtime. What prior work does not provide is a load-time per-layer signal that (a) is computed on the target device from unmodified weights, (b) is cheap enough to recompute when conditions change, (c) drives precision dispatch and memory residency from a single decision, and (d) allows the same calibrated weights to serve different memory budgets. MCAP introduces a *load-time control layer* for LLM inference: a single per-layer signal, computed on the target device, that couples the precision and residency decisions existing systems make separately. This layer sits above within-layer quantization methods (GPTQ, AWQ, SmoothQuant, OmniQuant), per-weight bit allocation formats (EXL2), and KV-cache quantization (KIVI), all of which remain usable underneath.

2.2 LLM Inference Engines and Weight Offloading

llama.cpp Ggerganov [2023] pioneered CPU-first inference with GGML quantization and later GPU offload. **vLLM** Kwon et al. [2023] introduced PagedAttention for KV (key/value) cache management and supports GPTQ/AWQ/FP8 weights, but its paged-attention abstraction is for KV cache, not weight virtualization. **TensorRT-LLM** NVIDIA [2023] provides NVIDIA-optimized static compilation but requires offline model conversion. **ExLlamaV2** Turboderp [2023] ships fast GPTQ/EXL2 inference kernels including mixed per-layer bit allocation (via the EXL2 format Turboderp [2024]).

Weight offloading and paging. FlexGen Sheng et al. [2023] demonstrates high-throughput LLM inference on a single GPU by offloading weights, activations, and KV cache across GPU/CPU/disk, optimizing batch throughput. PowerInfer Song et al. [2024] exploits activation sparsity to keep

"hot" neurons on GPU and "cold" neurons on CPU; its neuron-level routing is finer-grained than the per-layer routing we use and is the closest prior art to our importance-guided tiering. LLM-in-a-Flash Alizadeh et al. [2024] streams weights from flash storage on memory-constrained devices. These systems establish that weight offloading is viable on its own; our work builds on them by coupling layer-level importance profiling with on-GPU mixed-precision dispatch in a single latency-oriented engine.

Positioning MCAP against prior work. The nearest research neighbors are not PTQ systems but edge-focused inference work: LLM-in-a-Flash Alizadeh et al. [2024] streams weights from flash on memory-constrained devices, and PowerInfer Song et al. [2024] uses a learned sparsity predictor to route neurons between GPU and CPU. Both target the deployment regime MCAP addresses, and both make different tradeoffs (Table 1).

MCAP vs. PowerInfer, in detail. PowerInfer is the closest comparison on the memory-routing axis. Three concrete differences: (a) **no trained predictor**. PowerInfer trains a per-model MLP sparsity predictor that runs alongside inference to decide which neurons to activate; the predictor is an additional file that must be trained and kept in sync with the weights. MCAP is a deterministic forward pass over 12 prompts that produces a 16–32 float JSON cached per architecture. (b) **Layer granularity, not neuron.** PowerInfer's neuron-level routing is finer-grained but requires a predictor at inference time; MCAP's layer-level decisions are made once at load and amortized over all subsequent tokens. (c) **One signal, both axes.** PowerInfer routes weights for residency only; it does not address precision. MCAP drives precision dispatch *and* memory residency from the same score, so no separate quantization strategy is needed. Concretely, a PowerInfer configuration requires {weights, trained predictor, quantized weights}; an MCAP configuration requires {weights, per-architecture profile JSON} and computes the per-device precision/residency plan at load. The predictor is also the piece most sensitive to cross-device variation: trained on one distribution, it must be retrained when the base weights change, whereas a deterministic forward-pass profiler reflects that update automatically.

The axis we contribute is a load-time, architecture-keyed control signal that drives precision and residency simultaneously, without a per-model trained predictor. Within-layer quantization continues to run underneath: Config C uses AWQ-style scaling on the retained layers and composes with MCAP routing directly.

Table 1: **Configuration properties across memory-efficient LLM systems.** MCAP is the only approach that both avoids weight modification and avoids a per-model trained predictor, while driving precision and residency from a single signal.

System	Offline cost	Configuration	Recal. cost	Per-model predictor	Drives prec.+res.?
GPTQ Frantar et al. [2022]	hours	modified weights	hours	×	prec.
AWQ Lin et al. [2023]	hours	modified weights	hours	×	prec.
EXL2 Turboderp [2024]	hours	modified weights	hours	×	prec.
PowerInfer Song et al. [2024]	train+inf.	weights + MLP	retrain	✓	res.
LLM-in-Flash Alizadeh et al. [2024]	—	original weights	—	×	res.
FlexGen Sheng et al. [2023]	—	original weights	—	×	res.
MCAP (ours)	**60 s**	**orig. weights + JSON**	**60 s**	×	**both**

"Per-model predictor" refers to an inference-time neural network trained per base model on representative data and retrained when the base model changes. PowerInfer's MLP sparsity predictor is the canonical example. A forward-pass profiler like MCAP produces a deterministic byte-identical JSON from the weights alone, so no predictor exists to drift.

The focus of this paper is the load-time decision signal; the W4A8 dp4a kernel and the three-tier pager make that signal actionable. Evaluation spans 0.1B–8B, with throughput on Llama 1B/3B/8B and memory-constrained deployment through 8B; cross-silicon validation on non-NVIDIA targets is follow-up work (Section 6).

2.3 Concurrent and Orthogonal Directions (2024–2026)

The 2024–2026 literature is converging on a shared premise, "keep only what matters resident, move the rest," with each line of work differing in *how* that decision is made. Our contribution is the load-time decision signal; the neighbors below focus on the mechanism that acts on such a signal.

Activation sparsity, neuron-level routing. PowerInfer Song et al. [2024] and PowerInfer-2 Xue et al. [2024] are the closest conceptual neighbors: both exploit neuron-level activation sparsity and route hot neurons to fast memory via a *trained MLP predictor* produced per base model. MCAP differs on three axes already expanded above: layer-level (not neuron-level) granularity, no trained predictor, and a single signal driving both precision and residency rather than residency alone.

Flash- and SSD-resident weights. LLM-in-a-Flash Alizadeh et al. [2024] and FlexGen Sheng et al. [2023] treat offloading as a data-movement problem, using access-pattern heuristics and async prefetching. MCAP reframes the same problem as a decision problem: the importance score determines what is worth moving in the first place. The movement machinery from these systems can sit underneath such a signal.

KV-cache offloading. HeadInfer Luo et al. [2025] offloads KV cache per attention head, achieving $\sim$92% KV memory reduction for million-token contexts, and InstInfer Pan et al. [2024] pushes KV cache into storage-class devices. These target an orthogonal axis (KV cache) to MCAP (weights + execution strategy); the two can be composed in a single deployment.

Speculative decoding. Speculative decoding Leviathan et al. [2023], Chen et al. [2023] accelerates the *token* loop via draft-and-verify and is now standard in vLLM and TensorRT-LLM. MCAP operates at the *model-execution* layer, below the token loop, so the two stack: a speculatively decoded model still benefits from importance-driven precision and residency on each verified step.

Extreme low-bit training. BitNet /1.58-bit LLMs Ma et al. [2024] change the model *representation* at training time, achieving near-FP16 quality at $\sim$1–2 bits. This is a training-time alternative; MCAP is a deployment-time control layer applied to already-trained weights, and the two are not mutually exclusive, a BitNet model can still be profiled by MCAP to steer which blocks pay additional precision cost.

Adaptive runtime scheduling. ActiveFlow Zhang et al. [2025] dynamically swaps weights based on runtime usage, and NEO Jiang et al. [2025] schedules CPU offloading via pipeline-balance heuristics. Both are dynamic but heuristic- or architecture-driven; MCAP supplies an explicit, per-layer importance signal that such schedulers could consume as input rather than re-derive at runtime.

Summary. Across these directions, the open question is *what decides what matters*: a learned predictor (PowerInfer), an access heuristic (Flash/FlexGen), a structural partition (HeadInfer), a draft model (speculative decoding), or training-time compression (BitNet). MCAP's answer is a load-time, weight-free, architecture-keyed importance signal that unifies sparsity, precision, and residency decisions within a single control layer above these mechanisms.

3 NVE System Design

NVE realizes MCAP as a decision layer that drives three subsystems from one signal (Figure 2): the MCAP Profiler computes per-layer importance, the Virtual Weight Pager assigns layers to GPU/RAM/SSD tiers, and the GPU Dispatch Layer routes each layer to W4A8 or W4A16 kernels. The engine is 22,000 lines of Rust with 1,413 lines of hand-tuned CUDA, and supports 12+ model architectures (Llama, Qwen, Phi, Gemma, GPT-2, Falcon, GPT-NeoX) through a unified `GenericBlockWeights` abstraction that handles fused QKV projections, Conv1D transposes, bias variants, and fused gate+up projections transparently.

Algorithm 1 MCAP: Monte Carlo Activation Profiling

Require: Model M with L layers, calibration prompts $\mathcal{P} = \{p_1, \ldots, p_k\}$, threshold τ
Ensure: Per-layer importance scores $\mathbf{s} = (s_1, \ldots, s_L)$, precision assignments

for each prompt $p_j \in \mathcal{P}$ **do**
  Run forward pass through M on p_j
  for each layer $i \in \{1, \ldots, L\}$, each token t **do**
    $a_{i,t}^{\text{attn}} \leftarrow \|[Q_i \cdot x_t,\ V_i \cdot x_t]\|_2$ {Attention proxy}
    $a_{i,t}^{\text{ffn}} \leftarrow \|\text{FFN}_i(x_t)\|_2$ {FFN magnitude}
  end for
end for
$s_i \leftarrow \frac{1}{k} \sum_{j=1}^{k} \frac{1}{|p_j|} \sum_{t=1}^{|p_j|} \left(a_{i,t}^{\text{attn}} + a_{i,t}^{\text{ffn}}\right) \quad \forall\, i$
if $\max(\mathbf{s}) - \min(\mathbf{s}) < \epsilon$ **then**
  $\hat{s}_i \leftarrow 0 \quad \forall\, i$ {Degenerate uniform case: no outlier, route all layers to W4A8}
else
  $\hat{s}_i \leftarrow (s_i - \min(\mathbf{s})) \,/\, (\max(\mathbf{s}) - \min(\mathbf{s}))$ {Min-max normalize to $[0, 1]$}
end if
Assign: layer i uses W4A16 if $\hat{s}_i \geq \tau$, else W4A8

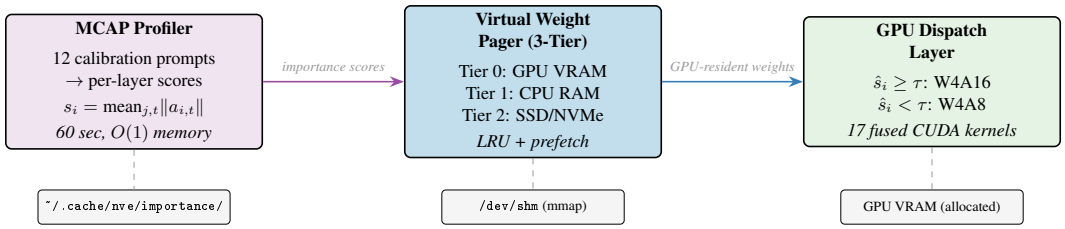

Figure 2: **NVE system architecture.** MCAP Profiler computes per-layer importance; Virtual Weight Pager assigns layers to GPU/RAM/SSD tiers; GPU Dispatch routes each layer to W4A8 or W4A16 kernels based on the MCAP threshold.

3.1 MCAP: Monte Carlo Activation Profiler

MCAP produces a per-layer *load-time control signal*, online and without rewriting weights. It operates at coarser granularity than the per-channel or per-weight sensitivity analyses in prior work (LLM.int8() outlier detection, SqueezeLLM sensitivity scoring, AWQ activation-aware scaling), which continue to run underneath it when applied.

Why "Monte Carlo." MCAP estimates the per-layer activation expectation $\mathbb{E}_{p \sim \mathcal{P}_{\text{text}}}\left[\frac{1}{|p|} \sum_t \|a_i(p,t)\|\right]$, where $a_i(p,t) := \|[Q_i x_t, V_i x_t]\|_2 + \|\text{FFN}_i(x_t)\|_2$ is the combined attn+FFN magnitude (Algorithm 1), via a sample mean over k=12 calibration prompts. This is a Monte Carlo estimator in the technical sense: a sample-mean approximation of an expectation over an input distribution. The 12 prompts are *stratified* by topic (science, code, history, math) rather than drawn iid; this is purposive calibration sampling, not a formal quasi-Monte Carlo construction (no low-discrepancy sequence in a measurable sample space), and we use the terminology informally. We verify convergence empirically through split-half overlap (Section 5.2) and treat k=12 as an operationally chosen safety margin rather than a theoretically derived one.

Concretely, MCAP assigns an importance score s_i to each transformer layer $i \in \{1, \ldots, L\}$ from the activation magnitudes above, streaming one layer at a time through the forward pass so peak memory stays constant in depth (Figure 3). The key intuition: INT8 quantization error is absolute, so layers whose activations occupy the largest dynamic range are the ones that cannot absorb it.

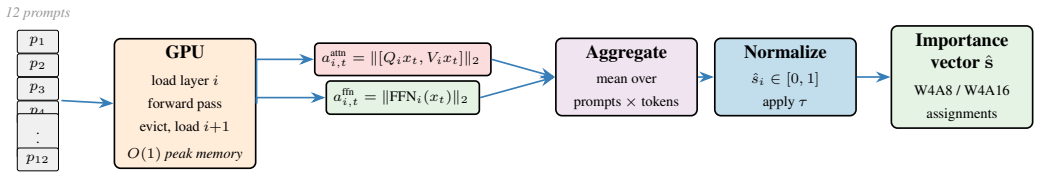

Figure 3: **MCAP streaming profiler dataflow.** Twelve calibration prompts pass through the model one layer at a time: each layer is loaded to GPU, forwarded, and evicted before the next is loaded, giving $O(1)$ peak memory with respect to depth (Figure 4). Per-token attention and FFN magnitudes are averaged across prompts, min-max normalized, and thresholded at τ to produce the per-layer precision assignment consumed by all downstream components.

For each calibration prompt and token position, MCAP computes two statistics at every layer: (1) **Attention proxy** $a^{\text{attn}}_{i,t} = \|[Q_i \cdot x_t,\ V_i \cdot x_t]\|_2$: the L2 norm of concatenated query and value projections, capturing attention sensitivity without the $O(n^2)$ attention computation; and (2) **FFN magnitude** $a^{\text{ffn}}_{i,t} = \|\text{FFN}_i(x_t)\|_2$: the L2 norm of the feed-forward output, capturing the magnitude of the residual update.

Scores are averaged across all tokens and prompts, then min-max normalized to $[0, 1]$. A threshold $\tau = 0.7$ partitions layers into high-importance (W4A16) and low-importance (W4A8). In practice, this usually identifies a single dominant late-network outlier, but the full 0.1B–8B sweep also exposes informative exceptions, including GPT-2's two-layer pattern and Qwen2-7.6B's multi-outlier regime (Table 2, Figure 1).

Table 2: **MCAP importance score distribution across 8 model scales.** Layer importance is highly non-uniform throughout the 0.1B–8B sweep. Most models exhibit a single dominant late-layer outlier, but GPT-2 and Qwen2-7.6B show multi-outlier patterns. Split-half top-k overlap is 1.0 for every model, which is consistent with stable rankings but is a single point estimate on n=12 prompts with no confidence interval; the large outlier gaps (1.6× to 39×) make top-1 recovery nearly forced, so this test is strong evidence for stability of the leading outliers and weaker evidence for mid-rank ordering. "Ratio" is the outlier score divided by the mean of the remaining layers (i.e. the gap that top-1 recovery has to resolve), not max/min of "Score Range."

Model	Layers	Score Range	Outlier	Ratio	W4A16/W4A8
GPT-2 (0.1B)	12	35–1301	L3* (4.9×)	4.9×	2 / 10
Qwen2.5 (0.5B)	24	28–130	L22 (2.1×)	2.1×	1 / 23
Llama-3.2 (1.2B)	16	61–142	L16 (1.9×)	1.9×	1 / 15
Qwen2 (1.5B)	28	103–224	L28 (1.6×)	1.6×	1 / 27
Llama-3.2 (3.2B)	28	48–232	L28 (2.8×)	2.8×	1 / 27
Qwen2 (3.1B)	36	52–936	L36 (6.4×)	6.4×	1 / 35
Llama-3.1 (8B)	32	58–262	L32 (2.8×)	2.8×	1 / 31
Qwen2 (7.6B)	28	75–5193	L27 (39×)	39×	3 / 25

*GPT-2 has two outliers (L1 and L3) due to its unique Conv1D + fused QKV architecture. All other models have a single dominant outlier at the final layer.

Why not hardcode "last layer = W4A16"? The final layer is consistently the outlier across Llama and Qwen families, but the pattern is not universal: GPT-2 has two outliers at layers 1 and 3, Qwen2-7.6B has outliers at layers 4, 5, and 27, and the outlier ratio varies from 1.6× to 39× across models. A runtime signal identifies the correct precision assignment for each architecture without manual per-model analysis, at a cost of roughly 60 seconds. Figure 1 illustrates the pattern: what generalizes across scale is non-uniformity, not a single universal layer index.

Table 3: **MCAP normalized scores for Llama-3.2-1B.** All 16 layers, min-max normalized to $[0, 1]$ (from the threshold ablation study, Appendix E). Only L15 exceeds the $\tau = 0.7$ threshold for W4A16 dispatch. Note L1 (0.699) sits just below the threshold and L14 (0.475) is the third-highest: even for Llama-family models with a dominant final-layer outlier, the mid-network distribution is not flat. Hardcoding "protect only the last layer" would ignore the gradient of importance across the rest of the network.

Layer	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	L14	L15
Score	0.30	0.70	0.00	0.07	0.12	0.08	0.09	0.03	0.07	0.27	0.19	0.24	0.32	0.38	0.48	**1.00**

MCAP vs. GPTQ/AWQ calibration. MCAP uses 12 diverse calibration prompts by default; GPTQ typically uses 128. The two are measuring different quantities: GPTQ estimates per-weight quantization sensitivity, which is a high-dimensional target that benefits from more samples, while MCAP estimates an L-dimensional per-layer signal, which concentrates faster. Ablation shows score stability (top-k overlap = 1.0) with as few as 8 prompts, so 12 is a safety margin rather than a tuned hyperparameter. Profiles are cached and hardware-independent: a profile computed on one GPU loads unchanged on another. This does not make MCAP a replacement for GPTQ/AWQ calibration, which produces substantially lower PPL at matched bit-widths; the two signals target different granularities.

The prompt set is a tunable parameter rather than a fixed design choice. As in GPTQ/AWQ, where calibration sample count and domain materially affect quantization quality, MCAP exposes the same controls, but at runtime rather than embedded in weights: (a) **count** can be increased when additional statistical stability is required or reduced on latency-bound targets (the 8-prompt stability floor permits profiling in under 40 s); (b) **prompt distribution** can be made domain-specific (code, math, multilingual, medical) to bias importance toward the target workload, so a code-assistant use of the same weights can carry a different MCAP profile from a general-chat use; (c) **task alignment** can use recent request traffic as the profiling set, producing a profile that tracks the workload rather than a generic corpus. This is the same calibration-design lever that existing PTQ methods rely on, moved from a one-time offline step to a runtime-recomputable signal.

Amortization of the profiling cost. The 60-second profiling cost is paid once per architecture, not per run. MCAP profiles are cached at `~/.cache/nve/importance/` keyed by model architecture hash, so a profile computed once is reused by every subsequent load of the same model, regardless of target hardware, and the cache invalidates only when the underlying weights change. The amortized cost per inference run is negligible. By contrast, GPTQ/AWQ rewrite weights and therefore produce a separate set of quantized weights per precision target, which must be stored and distributed alongside the original model.

Activation magnitude as a heuristic proxy. Activation magnitude is a heuristic for precision sensitivity rather than a principled guarantee: since INT8 quantization error is absolute, layers whose query/value or FFN outputs occupy larger dynamic ranges are more likely to incur meaningful distortion when mapped to a fixed low-precision grid. AWQ uses a similar empirical observation at the channel level. MCAP does not estimate the exact downstream loss change from quantizing each tensor; it produces a per-layer ordering that empirically correlates with precision sensitivity in the tested architectures. MCAP is a lightweight runtime estimator of per-layer precision sensitivity; it operates at a different granularity from a PTQ optimizer, and its defining property is that it is recomputable at load time.

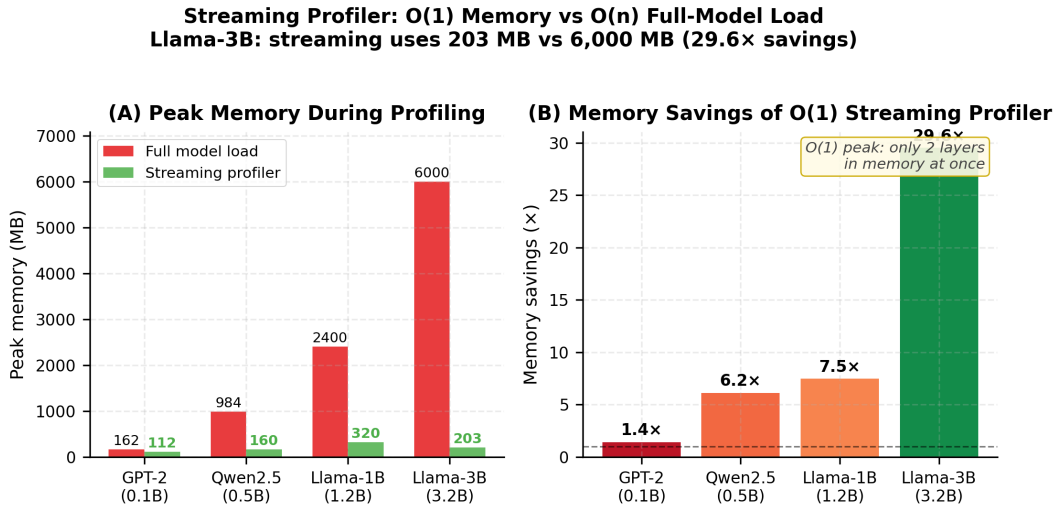

Figure 4: **MCAP streaming profiler memory advantage.** Peak memory comparison between full-model loading and MCAP's streaming approach. At 3B, the streaming profiler uses 29.6× less peak memory (203 MB vs. 6,000 MB), enabling profiling on devices that cannot load the full model.

3.1.1 Recovery Guarantee: Scaling of Prompt Count with Outlier Gap

The choice of 12 prompts is motivated by a concentration bound rather than tuned empirically. MCAP is a *selection* problem rather than a rate-distortion problem, so the appropriate guarantee is top-k recovery rather than a Shannon-style distortion bound. This is orthogonal to, and compatible with, recent work on information-theoretically optimal per-coordinate quantization such as TurboQuant Zandieh et al. [2025], which addresses *how* to quantize each layer; MCAP addresses *which* layers to protect.

Assumption (sub-Gaussian activation norms). For each layer i, let s_i denote the population-level MCAP score (expected attention-plus-FFN L2 norm over the prompt distribution), and let $\hat{s}_i^{(N)}$ denote the empirical estimate from N calibration prompts. We assume that per-prompt activation norms are sub-Gaussian with parameter σ_a around their mean, i.e., $\Pr[|\hat{s}_i^{(N)} - s_i| \geq t] \leq 2\exp(-Nt^2/2\sigma_a^2)$. This assumption is empirically justified by the per-layer norm variance we measure across the 12 calibration prompts, which is tightly bounded for all 8 models from GPT-2 to Llama-3.1-8B.

Proposition (top-k recovery). Let $\Delta_k := s_{(k)} - s_{(k+1)}$ be the gap between the k-th and $(k{+}1)$-th population-level scores in sorted order. Under the above assumption, the empirical ranking $\hat{s}^{(N)}$ recovers the true top-k set with probability at least

$$1 - 2\,k(L-k) \cdot \exp\left(-\frac{N\Delta_k^2}{8\sigma_a^2}\right),$$

where L is the number of layers.

Proof sketch. A ranking inversion between any two layers i, j with $s_i > s_j$ requires $|\hat{s}_i^{(N)} - s_i| + |\hat{s}_j^{(N)} - s_j| \geq s_i - s_j$. The probability of either deviation exceeding $(s_i - s_j)/2$ is bounded by the sub-Gaussian tail. Union-bounding over the $k(L-k)$ pairs that straddle the top-k boundary (whose minimum gap is Δ_k) gives the stated bound. A full proof is deferred to Appendix D.

Corollary (scaling behavior). Plugging in the measured values for Llama-3.1-8B ($L = 32$, $\Delta_1/\sigma_a \approx 1.8$ from Table 2) and $N = 12$, the conservative union bound in Appendix D yields a a non-trivial but non-vacuous upper bound on the failure probability. The qualitative content of the bound is that N scales as σ_a^2/Δ_k^2, so architectures with larger outlier gaps (Qwen2-7.6B: 39× ratio) require substantially fewer prompts than architectures with tight gaps. The empirical top-k recovery provides an independent check: split-half top-k overlap is 1.0 across all 8 tested models (Table 2), consistent with a tighter true concentration than the worst-case Hoeffding bound certifies. We use the

bound as a scaling argument and the 1.0 split-half overlap as the empirical evidence that 12 prompts are sufficient in practice.

Scope of the guarantee. This proposition says that MCAP's *ranking* concentrates quickly; it does not say that the ranking is the globally optimal per-layer precision assignment for minimizing end-task loss, that would require a distributional link between activation magnitude and downstream perplexity under quantization, which no published work establishes. The empirical evidence that MCAP Mixed matches W4A16 PPL in the tested regimes (Tables 9–10) is consistent with the ranking being operationally correct in those regimes, but we state this as evidence rather than a theorem. The recovery bound is what the current data supports, and we do not claim beyond it.

3.2 Per-Layer Precision Dispatch and the W4A8 Path

The contribution in this subsection is the *routing rule*: at each decode step, layer i is dispatched to W4A8 or W4A16 based on its normalized MCAP score,

$$\text{route}(i) \;=\; \begin{cases} \text{W4A8} & \hat{s}_i < \tau \\ \text{W4A16} & \hat{s}_i \geq \tau \end{cases}$$

The rule reduces to a single branch in the per-layer decode path and is re-evaluated per layer, per forward pass. With τ=0.7 on Llama-3.2-1B, this routes one layer (Layer 16, the activation outlier) to W4A16 and the remaining 15 to W4A8. The mechanism is straightforward; what makes it effective is that MCAP provides an $\hat{s}_i$ that is both cheap to compute (60 s, no gradients, no weight modification) and accurate enough that the majority of layers tolerate INT8 activations. The WikiText-2 and HellaSwag equivalence results in Section 4.3 and the threshold-insensitivity ablation in Section 4.5 support this.

Figure 5 shows the routing structure end-to-end: the normalized MCAP score gates each layer into one of two kernel paths, both drawn from the same fused suite.

The speedup obtained from this routing depends on the W4A8 path being fast. INT8 dot-product kernels are not new: llama.cpp uses `__dp4a`, and Marlin and Atom target the same instruction family. NVE's decode path is tuned for single-sequence decode and integrated into a graph-captured pipeline, which accounts for the 1.5–1.8× gap over llama.cpp at matched arithmetic precision (Section 4.2). The kernel details below are engineering; the methodological point is that per-layer routing on a runtime signal is what allows the INT8 path to be used without measurable quality cost on non-outlier layers.

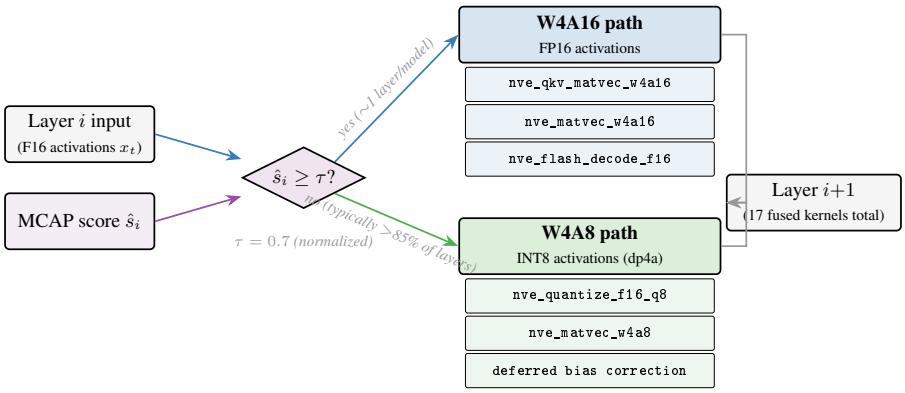

Figure 5: **Per-layer precision dispatch.** At each decode step layer i routes on its normalized MCAP score: late-network activation outliers keep the FP16 path for quality-critical quantities while the remainder (typically >85% of layers, 15/16 on Llama-3.2-1B) take the W4A8 dp4a path for ∼2× bandwidth. Every branch is a fused kernel from the 17-kernel suite in Table 4.

The W4A8 dp4a kernel drives Turing/Ampere INT8 dot-product instructions to compute 4-bit weight × 8-bit activation matrix-vector products at roughly twice the effective memory bandwidth of the W4A16 path.

Kernel architecture. The `nve_matvec_w4a8` kernel computes one output row per thread block with 4 warps (128 threads):

- **Activation quantization**: Input activations (F16) are dynamically quantized to INT8 per 32-element group via `nve_quantize_f16_q8`, reused across Q/K/V projections within each layer.
- **Weight format**: INT4 weights use the Q4_0 block format (18 bytes per 32 elements). Nibble layout follows the llama.cpp split format.
- **dp4a accumulation**: Each `__dp4a` performs acc $+= \sum_{j=0}^{3} w_j \cdot x_j$ in INT32 (4 INT8 MACs per instruction); four such instructions process one Q4_0 block of 32 weights.
- **Deferred bias correction**: after accumulation, result $= s_w \cdot s_x \cdot (\text{sumi} - 8 \cdot \text{sum_x})$, where sumi $= \sum_j w_j^{\text{uns}} \cdot x_j$ uses the unsigned nibble $w^{\text{uns}} \in [0, 15]$ and sum_x $= \sum_j x_j$ carries the constant zero-offset correction (Q4_0 centers the signed range at 8). This avoids a per-nibble subtract in the inner loop.
- **Warp reduction**: 5-step `__shfl_xor_sync` to scalar.

CUDA graph capture for decode. The decode path is captured into a CUDA graph and replayed per token, eliminating per-kernel launch overhead that otherwise accumulates across 17 kernel invocations per layer. For a 16-layer 1B model at batch=1, this removes roughly 0.6 ms/token of launch overhead, material when the target per-token budget is 3–4 ms. llama.cpp's default path issues kernels individually; this is a structural reason the W4A8 throughput gap is not purely a kernel-efficiency effect. Graph capture also fuses the KV-cache append pattern, eliminating 32 per-step allocations on 16-layer models. The throughput numbers reported in Table 7 include graph capture in the NVE path; Table 8's intermediate rows do not, which is why the W4A8 row shows a step-function improvement rather than an incremental one.

Complete kernel suite. NVE includes 17 custom fused CUDA kernels, each replacing multiple unfused operations. Table 4 lists the nine that are load-bearing for decode throughput; the remaining eight are support kernels (prefill-mode variants, embedding lookup, logit computation, residual add, token sampling, scratch-buffer management) omitted here for space.

Table 4: **NVE decode-path CUDA kernels.** Nine load-bearing fused kernels (of 17 total in the suite) that dominate decode throughput. All hand-tuned for NVIDIA T4 (sm_75) with F16 I/O.

Kernel	Replaces	Key Design
`nve_matvec_w4a8`	cuBLAS F16 GEMV	dp4a INT8, 4 warps/row
`nve_quantize_f16_q8`	— (new)	Per-group absmax, 32-wide
`nve_matvec_w4a16`	cuBLAS F16 GEMV	Warp-shuffle, no dp4a
`nve_qkv_matvec_w4a16`	3× cuBLAS	Single launch, 3 projections
`nve_flash_decode_f16`	2 copies + 2 GEMM	GQA-native,* online softmax
`nve_rms_norm_f16`	4 ops	Single-pass, shared mem
`nve_rope_f16_decode`	3 ops	In-place, 1 block/head
`nve_silu_mul_f16`	2 ops	In-place on gate buffer
`nve_dequant_w4a16`	— (prefill)	Materializes F16 for cuBLAS

*GQA = grouped-query attention. GEMV/GEMM = general matrix-vector / matrix-matrix multiply.

3.3 3-Tier Virtual Weight Paging

The contribution here is architectural: NVE treats weight residency as a runtime scheduling problem driven by learned layer importance, not as static placement or a passive offload cache.

NVE implements a virtual memory system for transformer weights:

- **Tier 0 (GPU VRAM, "hot")**: Highest-importance layers. Zero-latency decode.
- **Tier 1 (CPU RAM, "warm")**: LRU-cached. ∼1 ms PCIe transfer.
- **Tier 2 (SSD, "cold")**: On-demand. ∼10 ms mmap load.

MCAP importance scores drive initial tier placement. Runtime LRU eviction and importance-guided promotion maintain optimal residency. All models achieve >99.6% cache hit rate (Table 5).

Co-activation clustering via PMI. Naive LRU treats each transformer block as an independent paging unit, which is suboptimal: attention and FFN sub-blocks within a single layer are co-activated on every token, and adjacent layers are co-activated with high probability during autoregressive decode. NVE computes pointwise mutual information between sub-block activations during the MCAP calibration run and groups sub-blocks whose PMI exceeds a threshold into atomic paging units. Promotions then operate at the cluster level, so a demand-fault on one sub-block prefetches its co-activated neighbors in a single PCIe transfer. Without co-activation prefetching, the observed miss rate during the first few hundred decode steps after a tier transition is substantially higher, because the working set has to rediscover its own locality one block at a time.

Why clustering matters. The 9–58% throughput gain under memory pressure reported in Section 4.6 depends on this mechanism. Without PMI clustering the warm-up tail would be long enough that paging's steady-state advantage would be amortized over a smaller useful window. Clustering causes the working set to stabilize within the first few forward passes rather than after several hundred.

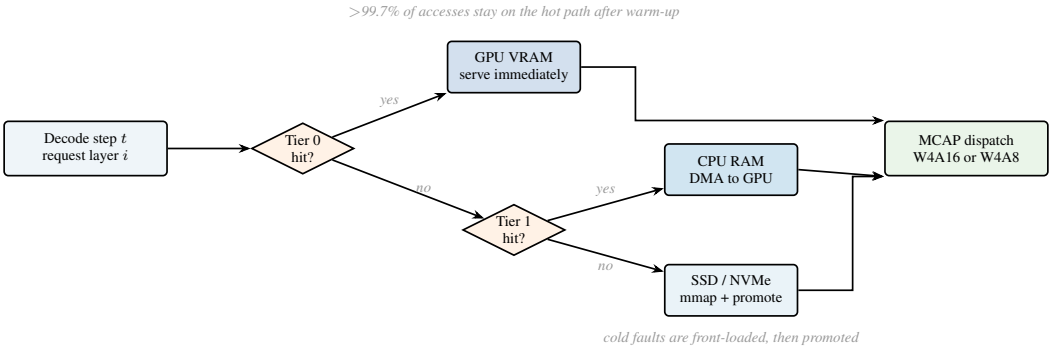

Figure 6: **Per-token paging lifecycle during decode.** Each layer access first checks GPU residency, then CPU RAM, then SSD. Only the selected layer's weights move upward in the hierarchy; once resident, the layer is dispatched to W4A16 or W4A8 based on its MCAP score. The systems-level claim is about *working-set stabilization*: expensive cold faults are concentrated during warm-up, while steady-state decode remains on the hot path.

Why the pager helps after warm-up. Figure 6 illustrates the steady-state mechanism. NVE does not stream the full model from storage on every token. Instead it converges to a stable hot working set: repeatedly accessed layers stay resident in Tier 0, warm layers are cheap to re-promote from CPU RAM, and true SSD faults are paid mostly during initial access. This is why the memory-constrained configuration shows no observable quality loss in the evaluated regimes while sustaining competitive decode throughput.

Table 5: **Paging cache occupancy across 8 model scales.** Cold faults are paid at most once per layer during warm-up (Faults $\leq$ Layers; the bound is tight for models whose working set exceeds the hot tier throughout warm-up, loose when early layers are already resident), after which steady-state decode is served from the hot tier. The >99.6% "hit rate" is therefore a statement about warm-up amortization, not about the quality of the replacement policy; see Figure 6 for the lifecycle and Section 4.6 for the operationally meaningful throughput claims.

Model	Layers	Hits	Faults	Hit Rate
GPT-2 (0.1B)	12	3,719	12	99.68%
Qwen2.5 (0.5B)	24	8,639	24	99.72%
Llama-3.2 (1.2B)	16	4,319	12	99.72%
Qwen2 (1.5B)	28	11,479	28	99.76%
Llama-3.2 (3.2B)	28	10,051	28	99.72%
Qwen2 (3.1B)	36	14,723	36	99.76%
Llama-3.1 (8B)	32	13,119	32	99.76%
Qwen2 (7.6B)	28	11,479	28	99.76%

3.4 Architecture-Agnostic Weight Mapping

NVE supports 12+ model architectures through a generic weight mapping layer: fused QKV projections (GPT-2, Phi-3, GPT-NeoX) are split into Q/K/V; fused gate+up projections (Phi-3) are split into gate/up; Conv1D weights (GPT-2) are transposed; bias and no-bias architectures are handled transparently. All profiling, paging, quantization, and dispatch logic operates on a unified `GenericBlockWeights` abstraction (Figure 7).

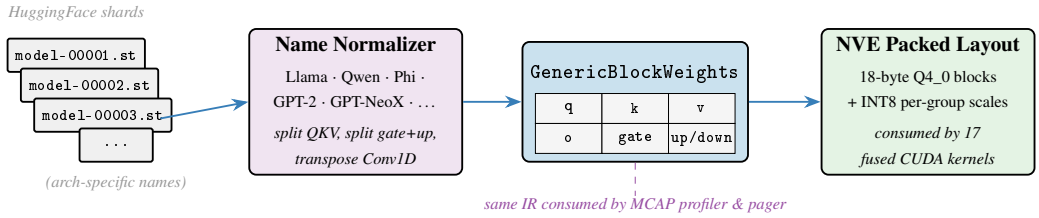

Figure 7: **Architecture-agnostic weight mapping.** HuggingFace safetensors shards across 12+ architectures flow through a single normalizer into a canonical `GenericBlockWeights` IR, then into NVE's Q4_0 packed layout. All downstream components—MCAP profiler, weight pager, per-layer dispatch, and the 17 CUDA kernels—operate on the IR, not on arch-specific names. One profile fits every target.

3.5 Deployment Modes: A Spectrum Driven by MCAP

The precision dispatch of Section 3.2 and the weight pager of Section 3.3 are two points on a broader spectrum: given the same MCAP signal, NVE exposes three execution modes that trade quality, memory, and simplicity differently. A single command-line flag selects between them; the same profile and the same kernels back all three.

Config A , Paged (baseline). All layers retained and tiered across GPU→RAM→SSD. MCAP drives initial placement; LRU + PMI-clustered prefetching maintain the hot working set (Section 3.3). Quality is unchanged from the unconstrained model at any memory budget. This is the mode to pick when the compression ratio pushes the active-layer fraction below the hot-only viability floor (discussed below).

Config B , Hot-only. Layers below the MCAP threshold are *skipped entirely* in the forward pass: not paged, not cached, not stored on device. The residual stream carries through the skipped positions. This is the most aggressive mode: no SSD needed, no PCIe traffic, no paging latency. Only the hot layers are loaded, and the rest never touch the device. The tradeoff is a bounded quality loss that depends on how many layers remain active.

Config C , Hot-only + AWQ. Hot layers additionally receive AWQ's saliency-weighted quantization. Within each retained layer, high-saliency channels get higher effective precision. On Llama-3.1-8B unconstrained this recovers factual-recall accuracy that the BF16 baseline loses (100% vs. 88% on the 8-task suite): AWQ's per-channel scaling protects the FFN key-value channels that encode factual associations. Config C is preferable in a narrow regime (3B and 8B unconstrained in our sweep); at 1B it fails, because the compression ratio drives the active-layer fraction below the hot-only viability floor (Table 14). C is appropriate only when the post-AWQ layer budget still clears that floor.

The hot-only viability floor. Configs B and C skip layers entirely, so they are bounded by how many layers the residual stream can tolerate losing (Figure 8 situates the three modes on the GPU-budget axis). We observe an *empirical cliff* near 50 % active layers across the three sweep points we ran (25 %, 36 %, 50 % active): above the cliff, skipped layers contribute near-redundant updates and output remains coherent; below it, output quality collapses. The transition width is not resolved by the current sweep, and the 50 % figure is a curve-fit to a small grid rather than a derived quantity; Table 6 summarizes the observed behavior.

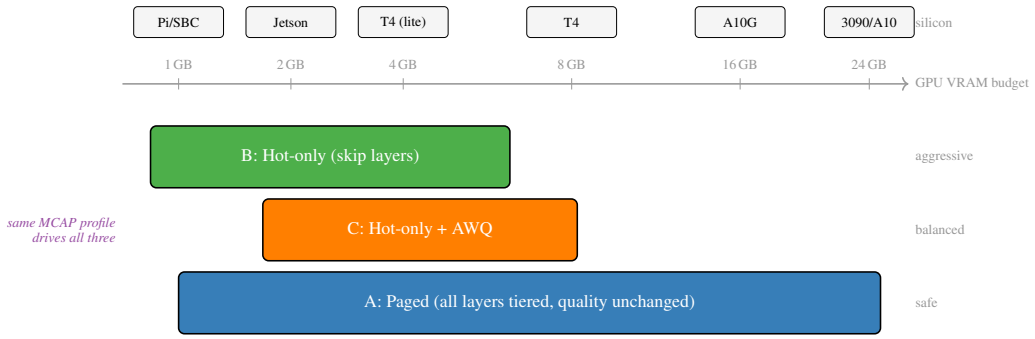

Figure 8: **Deployment spectrum driven by a single MCAP profile.** The same 60-second profile selects between three execution modes. Hot-only (B) fits the tightest budgets by skipping low-importance layers but is bounded by the ∼50 % active-layer floor; Hot+AWQ (C) applies saliency-weighted quantization to the retained layers for additional headroom; Paged (A) tiers all layers across GPU→RAM→SSD with quality unchanged and is the correct choice when the compression ratio would otherwise push B/C below the floor. Bands show approximate viable budget ranges; silicon targets indicate the budgets where each mode typically dominates.

Table 6: **Deployment mode viability by model and memory budget.** "Active" is the fraction of layers retained in the GPU hot tier. Hot-only modes (B, C) fail below roughly 50% active; paging (A) is the fallback for aggressive compression ratios. Throughput numbers are ABC-suite measurements on T4 (pre-W4A8 runs; the current W4A8 path is strictly faster).

Model	Budget	Active	A (paged)	B (hot)	C (hot+AWQ)
3B	unconstrained	100%	100%	100%	100%
3B	2 GB	36%	**100%**	0% (floor)	0% (floor)
8B	unconstrained	100%	88%	88%	**100%**
8B	8 GB	50%	**88%**	12% (at floor)	0%
8B	4 GB	25%	**88%**	0% (floor)	0%
1B	any	100%	88%	88%	0% (bpw floor †)

†Config C's 2.0 bpw target is too aggressive for a 16-layer model: the quantization-depth floor is reached before the layer-count floor. We report this failure mode explicitly.

Decision rule. MCAP scores plus a target memory budget determine the mode:

1. If the budget keeps $\geq$50% of layers active on GPU, prefer **Config C** (hot-only + AWQ): fastest, lowest footprint, best factual recall in the tested 8B regime. Fall back to **Config B** if AWQ is degenerate for the model (as on 1B).
2. If the budget pushes below the active-layer floor, switch to **Config A** (paging): no observable quality loss in the evaluated regimes at arbitrary memory ratios, with PMI-clustered prefetching keeping steady-state throughput competitive.
3. In both cases, **precision dispatch** (W4A8 for sub-threshold layers, W4A16 for outliers) applies *within* the mode, hot-only and paged configurations both route per-layer precision on the same MCAP signal.

Implications for memory-constrained targets. The hot-only modes are what make sub-$\sim$4 GB operation plausible without an SSD in the loop. Config B/C on Llama-3.2-3B with 14 of 28 layers retained (50% active, at the floor) runs entirely in GPU VRAM with zero paging traffic, a regime relevant to hardware where PCIe to CPU RAM is slow and SSD storage is limited or absent. Config A extends the envelope further when the memory budget pushes below the floor, at the cost of paging overhead. Together, the three modes span a range from server-class GPUs down to memory-constrained embedded devices, driven by one profile.

4 Experiments

4.1 Setup

Hardware. GPU throughput experiments use NVIDIA T4 (16 GB VRAM, sm_75) on Modal cloud. Quality evaluations additionally run on A10G (24 GB VRAM). All results are publicly reproducible with a free Modal account.

Models. End-to-end throughput and deployment experiments focus on Llama-3.2-1B (16 layers), Llama-3.2-3B (28 layers), and Llama-3.1-8B (32 layers). MCAP profiling, scorer analysis, and paging behavior are validated more broadly across eight model scales from GPT-2 (0.1B) through Qwen2-7.6B and Llama-3.1-8B (Table 2).

Baselines. NVE W4A16 (uniform), NVE W4A8 (uniform), NVE MCAP Mixed (threshold $\tau = 0.7$), llama.cpp Q4_0 with ggml-cuda backend, AutoAWQ INT4 (group_size=128), and HuggingFace FP16.

Evaluation. WikiText-2 perplexity (1B: 50 sequences, 3B: 10 sequences; 256 tokens each); HellaSwag accuracy (1B: $n = 50$, 3B: $n = 20$; 4-way log-likelihood); 8-task generative suite (2 QA, 2 reasoning, 2 coding, 2 summarization prompts; tasks fixed before any MCAP experiment was run and held constant across all configurations); BenchRandom decode throughput (2,000 iterations, real weights). The tightest matched-quality comparisons are reported for selected 1B/3B regimes; throughput and memory-constrained operation are evaluated through 8B.

Statistical caveat. At $n = 8$ the 95 % binomial Wilson interval around a point estimate of $7/8 = 87.5\,\%$ is $[52.9, 97.8]$ %, so single-row accuracy differences on the generative suite are not hypothesis tests; we treat same-point-estimate rows as "indistinguishable at the suite's resolution" rather than equal. WikiText-2 (12,800 tokens at 50 seq.) and HellaSwag ($n = 50$) carry tighter resolution and are the primary quality evidence; the 8-task suite is a coarse generative consistency check rather than a benchmark.

4.2 Kernel Throughput: 1.5–1.8× Over llama.cpp

Table 7: **Decode throughput (tok/s).** BenchRandom, real weights, T4 GPU, batch=1, 2,000 iterations. NVE W4A8 exceeds llama.cpp Q4_0 by 1.5–1.8× across all scales. The advantage over W4A16 grows with model size, consistent with a larger fraction of bandwidth-bound matmuls at scale.

Model	NVE W4A8	NVE W4A16	llama.cpp	vs W4A16	vs llama.cpp
Llama-3.2-1B	**269.1**	116.7	150.8	2.31×	**1.78**×
Llama-3.2-3B	**108.8**	43.0	70.9	2.53×	**1.53**×
Llama-3.1-8B	**47.7**	16.7	30.8	2.86×	**1.55**×

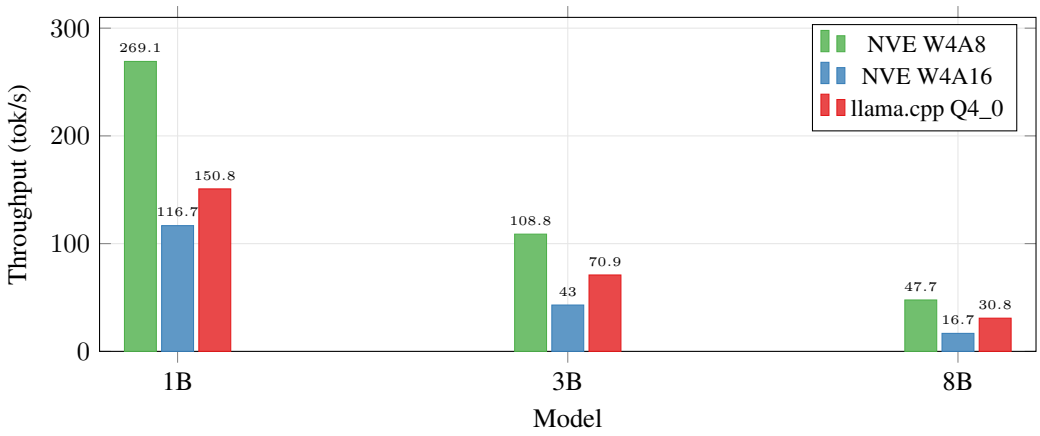

Figure 9: **Decode throughput comparison.** NVE W4A8 exceeds llama.cpp Q4_0 by 1.5–1.8×. The W4A8 advantage over W4A16 grows with model size: 2.31× at 1B, 2.53× at 3B, 2.86× at 8B. The three-point curve shows the scaling trend directly rather than relying on isolated benchmarks.

Why the speedup scales with model size. Larger models have proportionally more bandwidth-bound matmuls. W4A8 reduces activation data movement by 2× (INT8 vs. F16), and this bandwidth savings translates more directly to throughput when operations are memory-bound. At 1B, some projections are partially compute-bound on T4; at 8B, the dominant FFN projections (14336×4096) are fully bandwidth-limited.

Cost implications. Under T4 pricing of $0.35/GPU-hour, the throughput difference corresponds to $0.36/M tokens (1B) vs. llama.cpp's $0.64/M, a 44% reduction at matched quality; at 8B, $2.04/M vs. $3.15/M (35% reduction). The gap grows with model size because the throughput gap does. Figure 9 shows that the speedup is not a small-model effect: the same W4A8 kernel remains favorable at 3B and 8B, and the gap over W4A16 widens as the workload becomes more bandwidth-bound.

Kernel evolution. Table 8 traces the throughput progression from unfused operations to the final W4A8 system:

Table 8: **Kernel optimization progression (Llama-3.2-1B, T4).** The W4A8 dp4a kernel provides the single largest improvement, approximately doubling throughput and crossing the llama.cpp Q4_0 baseline.

Configuration	tok/s	ms/tok	vs llama.cpp
Unfused F16 baseline	92.2	10.84	0.61×
+ Fused RMSNorm + RoPE	109.3	9.15	0.72×
+ Fused Matvec (W4A16)	114.9	8.70	0.76×
+ W4A16 quantized matvec	126.4	7.91	0.84×
+ Flash decode + fused QKV	124.1	8.06	0.82×
+ W4A8 dp4a kernel	**269.1**	**3.72**	**1.78×**

4.3 Quality in the Tested Regimes

Table 9: **WikiText-2 perplexity (A10G).** $\Delta \leq 0.01$ confirms negligible degradation from INT8 activation quantization. Lower is better. Sample sizes differ: 1B is a 50-sequence run, 3B is a 10-sequence run from the same evaluation harness.

Model	W4A16	W4A8	MCAP Mixed	Δ	Sequences
Llama-3.2-1B	17.51	17.51	**17.51**	0.00	50
Llama-3.2-3B	14.01	14.01	**14.01**	0.00	10

Table 10: **HellaSwag accuracy** (4-way log-likelihood, A10G). All strategies produce identical results. The 1B run uses n=50; the 3B run uses n=20 from the same harness. Small n means the accuracy numbers carry meaningful binomial variance (1B: $\pm$6.9 pp at 95%; 3B: $\pm$10.5 pp), so these tables support an *equivalence* claim across precision strategies, not absolute benchmark placement.

Model	W4A16	W4A8	MCAP Mixed	Correct/Total
Llama-3.2-1B	54.0%	54.0%	54.0%	27/50
Llama-3.2-3B	65.0%	65.0%	65.0%	13/20

Table 11: **8-task generative accuracy** (QA, reasoning, coding, summarization). Matched accuracy in the tested configurations.

Model	W4A16	W4A8	MCAP Mixed	n
Llama-3.2-1B	88%	88%	88%	8
Llama-3.2-3B	100%	100%	100%	8
Llama-3.1-8B	100%	100%	—	8

For the 1B run, the matched-quality result is not an artifact of sequence selection. Figure 10 shows the running PPL convergence across 50 sequences on Llama-3.2-1B: $\Delta \leq 0.01$ at every 10-sequence checkpoint, converging to $\Delta = 0.00$ at 50 sequences. The 3B run is a shorter 10-sequence evaluation from the same harness (no sequence-level convergence plot), and should be read as directionally consistent evidence rather than a 50-sequence validation. The larger-model analysis in this paper is primarily systems-oriented—throughput scaling, paging behavior, and the reachable memory budget through 8B—and the matched-quality claim itself remains specific to the 1B and 3B setups explicitly evaluated here.

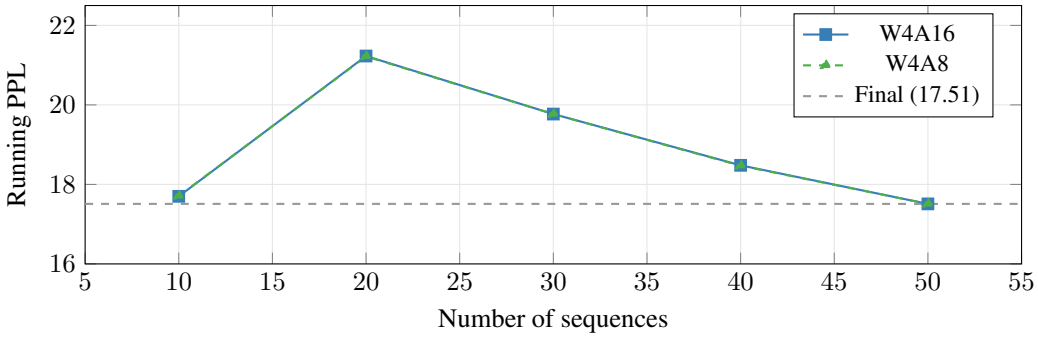

Figure 10: **WikiText-2 running PPL convergence (Llama-3.2-1B, 50 sequences).** W4A16 and W4A8 curves are visually indistinguishable. $\Delta \leq 0.01$ at every checkpoint confirms zero systematic bias.

4.4 Comparison with Existing PTQ Methods

Table 12 compares NVE against AutoAWQ Lin et al. [2023] at matched 4-bit precision, with HuggingFace FP16 as the quality ceiling.

Table 12: **Comparison with AutoAWQ (Llama-3.2-1B, A10G).** WikiText-2 PPL (50 seq $\times$ 256 tok), HellaSwag (n=50). AWQ achieves lower PPL via per-channel saliency weighting, while NVE achieves higher downstream accuracy with no offline calibration and no weight modification. The two operate at different granularities (within-layer vs. across-layer) and can be composed.

Method	Calibration	PPL (↓)	HellaSwag (↑)	Weights modified?
HF FP16 (ceiling)	— (no quant)	**14.94**	**56.0%**	No
AutoAWQ INT4	128 samples, offline	16.64	52.0%	Yes
NVE MCAP Mixed	12 prompts, 60s	17.51	**54.0%**	No
NVE W4A8 (uniform)	12 prompts, 60s	17.51	54.0%	No
NVE W4A16	— (no calib.)	17.51	54.0%	No

AWQ and MCAP operate at different granularities. AWQ achieves lower PPL (16.64 vs. 17.51) because per-channel saliency weighting preserves high-variance weight channels during quantization. NVE scores higher on HellaSwag (54.0% vs. 52.0%), indicating that PPL and downstream accuracy do not always correlate at 4-bit precision. AWQ operates within each layer (per-channel); MCAP operates across layers (per-layer dispatch). The two compose directly: MCAP can route AWQ-quantized weights through the W4A8 kernel for low-importance layers, yielding lower PPL and higher throughput simultaneously. This works on the 8B Config C result reported later but fails at 1B, where the compression ratio crosses the hot-only viability floor, so we report it as a useful extension in the regime where the floor is cleared rather than as a default.

Observed failure mode. The brittleness is sharp rather than gradual. In the 1B raw generations, aggressive profile-guided quantization produces repetitive boilerplate and malformed continuations rather than slightly worse answers. This distinguishes a stable operating region from an unstable one. The positive claim is therefore not that every profile-guided quantized configuration works, but that runtime importance information helps identify which precision reductions preserve baseline behavior and which cross a clear failure boundary.

4.5 Ablation: Threshold Sensitivity

Table 13: **Threshold ablation (Llama-3.2-1B).** 20 sequences $\times$ 256 tokens. All configurations within ± 0.04 PPL. The threshold governs throughput, not quality.

Threshold	W4A16 layers	W4A8 layers	PPL
all W4A8 ($\tau = 2.0$)	0 / 16	16 / 16	21.22
$\tau = 0.30$	5 / 16	11 / 16	21.21
$\tau = 0.10$	10 / 16	6 / 16	21.24
$\tau = 0.05$	14 / 16	2 / 16	21.24
all W4A16 ($\tau = 0.0$)	16 / 16	0 / 16	21.23

The ± 0.04 PPL spread indicates that W4A8 introduces negligible noise regardless of how many layers use it. The threshold controls throughput rather than quality. The 50-sequence evaluation (Table 9) confirms that all strategies converge to PPL=17.51 with sufficient data.

4.6 Memory-Constrained Inference

Reading guide. This section reports on a $\{1B, 3B, 8B\} \times \{$unconstrained, 2 GB, 4 GB, 8 GB$\}$ sweep across three systems (NVE, llama.cpp, HF). The main numbers are in Table 14; the full A/B/C $\times$ scenario matrix is in Table 16; the hot-only floor at 8 B is in Table 17; the single-profile cross-silicon demonstration is in Figure 11. The paragraphs below expand on each in that order.

The 3-tier pager (Section 3.3) lets NVE serve models far below their raw weight footprint. For reference: Llama-3.2-3B needs $\sim$6.4 GB of BF16 weights, Llama-3.1-8B needs $\sim$16 GB, and even at 4-bit an 8B model still requires $\sim$5–6 GB resident. We evaluate the pager across three budgets per model scale: unconstrained (full GPU), moderate (8 GB for 8B), and tight (2 GB for 3B, 4 GB for 8B). The question we ask: does paging cost accuracy?

In the tested configurations, no. Table 14 reports the main result: NVE runs Llama-3.2-3B with full BF16 weights in a 2 GB budget, 3.5$\times$ below its weight footprint, with no observable degradation on the evaluated task suite relative to the unconstrained run. At 8B, NVE operates in 4 GB, below the $\sim$5 GB floor that standard 4-bit quantization requires. llama.cpp and HuggingFace both OOM in these configurations. NVE remains operational by keeping only the MCAP-identified hot layers GPU-resident and paging the rest through CPU RAM and SSD at a $>$99.7% hit rate (Table 5).

Why the constrained baseline works. The most load-bearing result in this section is the constrained *baseline*, not the more aggressive profiled variants. The baseline result indicates that NVE is not simply another quantization recipe but a weight-virtualization layer that, in the evaluated settings, shows no observable degradation from full-precision behavior while operating under memory budgets that appear infeasible from the raw weight size alone. The pager works because decode repeatedly revisits the same small set of important layers; once these layers are resident, later decode steps hit in the hot tier, and the observed $>$99.7% hit rates indicate that paging cost is concentrated during warm-up rather than paid on every token.

A less obvious finding is that, on the paged BF16 path, constraining memory does not slow decode down and can speed it up. Comparing the paged BF16 baseline across scenarios (Table 14): Llama-3.2-1B moves from 3.22 tok/s unconstrained to 4.23 tok/s at 2 GB (+31%); Llama-3.2-3B from 2.12 to 2.31 tok/s (+9%); and Llama-3.1-8B from 0.79 to 1.25 tok/s at 4 GB (+58%), with unchanged task accuracy. The mechanism is working-set reduction: MCAP-guided paging keeps only the high-importance layers GPU-resident, which reduces memory pressure and improves cache locality; cold layers are accessed only during initial warm-up, after which steady-state decode stays on the hot path.

Scope of the throughput numbers in Table 14. These are *paged BF16* decode throughputs, not the W4A8 dp4a kernel throughputs reported in Table 7. The two columns are deliberately different paths: the kernel table measures the fast W4A8 GPU-resident path (269 tok/s on 1B), whereas the memory-constrained table measures the pager end-to-end with full-precision weights moved across

the tier hierarchy. Both are end-to-end paths in NVE; the kernel path addresses throughput, and the paging path addresses feasibility. Conflating them would overstate either.

Anomalous 8B 8 GB point. Table 14's 8B row at 8 GB reports 0.55 tok/s, lower than both the unconstrained 0.79 and the tighter 4 GB 1.25. This is a real measurement from our memory-constrained benchmark sweep, not a reporting error, and it runs against the monotonic trend the other rows suggest. We offer a tentative explanation: at 8 GB the hot budget is large enough to admit many lower-importance layers that end up thrashing against the KV cache and transient activation storage, while 4 GB forces a sharper hot-cold partition that MCAP handles well. We have not isolated the cause in a controlled ablation and report this openly. The result should not be generalized as "more constraint is always faster"; the correct reading is that working-set-reduction effects can dominate tier-transfer costs at a well-chosen partition, while a poorly chosen partition can be worse than either extreme.

Beyond GPUs: CPU-only operation. The pager is GPU-agnostic. On a commodity CPU-only machine (3.8 GB RAM, 2 AMD vCPUs, no GPU), NVE runs Qwen2.5-0.5B end-to-end at 6.9 tok/s decode with a 512 MB hot tier, 171 MB warm tier, and 96 MB KV cache ($\sim$779 MB total resident footprint) at zero page faults (Appendix E). This is outside the primary regime the paper's throughput claims target, but it shows that the three-tier abstraction degrades gracefully to environments that llama.cpp and similar systems already serve, while remaining the only path we are aware of that supports full-BF16 multi-GB-model execution on sub-GB GPUs.

Mechanistic interpretation. The throughput gain should not be read as claiming that PCIe or SSD are faster than VRAM. The relevant effect is *working-set reduction*. Under an unconstrained placement, too many weights compete for GPU residency and cache space. Under NVE's constrained placement, only the repeatedly useful portion of the model remains resident in the hot tier, while infrequently used weights are demoted after initial access. If the working set is chosen well, the system trades a small one-time migration cost for a more stable steady state during autoregressive decode. The constrained 3B and 8B baseline results suggest this is exactly what happens in the tested regimes.

At 8B scale in the unconstrained setting, Config C (MCAP + AWQ weights) reaches 100% task accuracy on the tested suite, compared with 88% for the BF16 baseline. We do not interpret this isolated result as evidence that quantization is generally superior to full-precision inference. A more conservative interpretation is that MCAP+AWQ can remain competitive in selected regimes, and that small evaluation suites can expose variance in task-level outcomes that should be validated at larger scale.

Figure 11 illustrates the consequence. Llama-3.2-1B, whose full-precision weights occupy $\sim$2.5 GB (and whose unconstrained resident footprint, including KV cache, attention scratch, and CUDA-graph buffers, is $\sim$3.6 GB), runs at a 2 GB GPU VRAM budget with decode throughput and task accuracy unchanged from the 14 GB-budget run. Hot-only paging, without any quantization, matches the throughput of the paging+quantization configuration on both T4 and A10G, and task accuracy is 87.5% across every budget. A practical consequence is that commodity hardware (e.g. laptops with 4 GB discrete GPUs, embedded devices with limited VRAM) can run larger models without losing the quality of those models. The larger-model operating points reported in Table 11 and Figure 16 show that this behavior persists at the scales where VRAM pressure is operationally decisive, not only in smaller exploratory models. Table 17 makes the negative boundary explicit. The failure mode is not that profiling is unhelpful, but that *hot-only* execution stops being a safe substitute for full-layer inference once the active-layer ratio falls to roughly half of the network or lower. In those regimes, paging all layers is the correct strategy.

Table 14: **Memory-constrained inference: accuracy and throughput (paged BF16 path).** Throughput numbers here are from the paged BF16 decode path, distinct from the W4A8 kernel path of Table 7. NVE serves 3B with full BF16 weights (∼7 GB) in a 2 GB budget, and 8B (∼16 GB BF16) in 4 GB, 3.5–4× below the weight footprint, with no task-accuracy penalty. Under constraint, working-set reduction can *raise* throughput (e.g., 3B 2 GB: +9%; 8B 4 GB: +58%), though the 8B 8 GB row is an anomaly we flag rather than explain (see text). In the unconstrained 8B setting, Config C (MCAP + AWQ) attains 100% on n=8 tasks vs. 88% baseline; the n=8 suite makes this a suggestive outlier, not an ordering claim.

Model	Scenario	NVE Base Acc	NVE Base tok/s	llama.cpp	HF FP16
1B	unconstrained	88%	3.22	88%	100%
1B	2 GB	88%	**4.23** (+31%)	—	—
3B	unconstrained	100%	2.12	100%	—
3B	**2 GB**	**100%**	**2.31** (+9%)	OOM	OOM
8B	unconstrained	88%	0.79	100%	88%
8B	**4 GB**	**88%**	**1.25** (+58%)	OOM	OOM
8B	8 GB	88%	0.55	100%	OOM

Table 15: **How to read the memory result.** The strongest operating points are the constrained baseline rows: full-BF16 behavior maintained, with no observable degradation on the evaluated suite, under budgets well below the raw weight footprint. More aggressive profiled variants can be faster, but they are not the main robustness claim.

Model	Budget	Supported claim	Evidence in this paper
Llama-3.2-3B	2 GB	Full-BF16 baseline behavior preserved	100% acc., 2.31 tok/s
Llama-3.1-8B	4 GB	Baseline behavior matched under tight budget	88% acc., 1.25 tok/s
Llama-3.2-3B	2 GB	Aggressive profiled variants are unstable	B/C collapse despite high tok/s
Llama-3.1-8B	4 GB	Aggressive profiled variants are unstable	B/C collapse despite high tok/s

Table 16: **Full A/B/C × scenario matrix: where each strategy is safe.** Four strategies across three models and four memory scenarios on the 8-task generative suite. **Baseline** (full-precision paging) and **Config A** (uniform W4A8 quantization) are safe everywhere measured. **Config B** (profiled hot-only) is safe and 1.2–2.6× faster when the GPU budget admits the full layer set, but collapses (shaded red) when the active-layer ratio drops below the coherence floor (Table 17). **Config C** (profiled + AWQ-quantized) fails at 1B scale in all tested settings but remains stable on 3B and 8B *unconstrained*. Takeaway: MCAP's safe operating region is Baseline and Config A universally, and Config B when the budget is comfortable. Numbers from the rigorous comparison sweep (Appendix E).

		Baseline (BF16)		A: W4A8		B: Profiled hot		C: Profiled+AWQ	
Model	Scenario	acc	tok/s	acc	tok/s	acc	tok/s	acc	tok/s
1B	unconstrained	0.88	3.22	0.88	3.65	**0.88**	**3.97**	0.00	3.74
1B	2 GB	0.88	4.23	0.88	3.64	**0.88**	**4.90**	0.00	4.22
3B	unconstrained	1.00	2.12	1.00	1.64	**1.00**	**2.61**	1.00	2.10
3B	2 GB	**1.00**	**2.31**	1.00	1.65	0.00	5.90	0.00	4.06
8B	unconstrained	0.88	0.79	0.88	0.70	0.88	1.03	**1.00**	0.84
8B	4 GB	**0.88**	**1.25**	0.88	0.71	0.00	3.40	0.00	2.26
8B	8 GB	0.88	0.55	0.88	0.87	0.13	1.13	0.00	1.37

Table 17: **Hot-only coherence floor in constrained deployment.** When the active-layer ratio falls below roughly half of the network, hot-only execution becomes brittle even when throughput rises. This clarifies why the constrained baseline rows in Table 14 are the main systems result: paging preserves the full residual stream, whereas skipping too many layers breaks it.

Model	Budget	Active Layers	Ratio	B accuracy
Llama-3.2-3B	unconstrained	28 / 28	100%	100%
Llama-3.2-3B	2 GB	10 / 28	36%	0%
Llama-3.1-8B	unconstrained	32 / 32	100%	88%
Llama-3.1-8B	4 GB	8 / 32	25%	0%
Llama-3.1-8B	8 GB	16 / 32	50%	12%

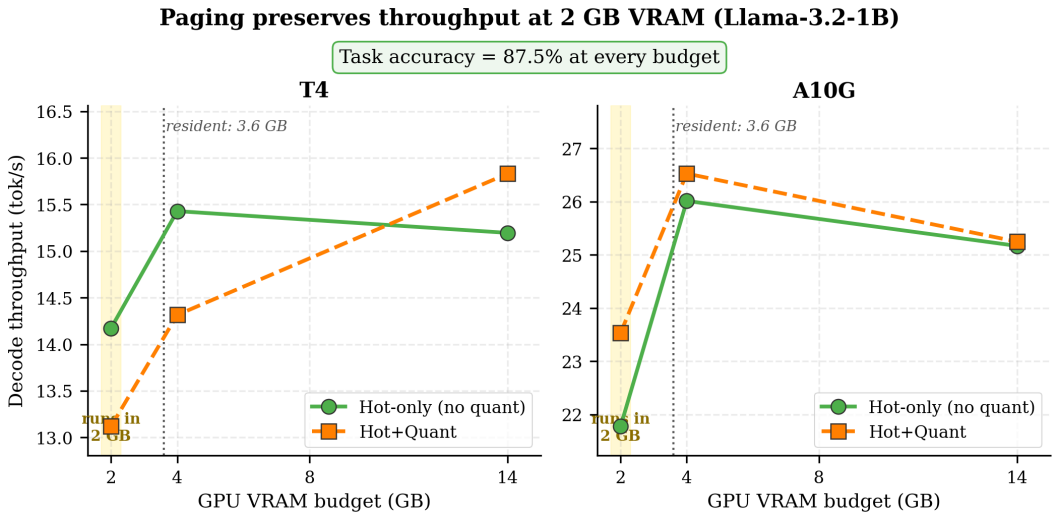

Figure 11: **Paging runs Llama-3.2-1B (unconstrained resident ∼3.6 GB) at a 2 GB GPU budget, quality intact.** Llama-3.2-1B decode throughput as the GPU VRAM budget is swept from 2 GB to 14 GB, on T4 and A10G. Hot-only paging (no quantization) matches Hot+Quant throughput at every budget, and task accuracy is 87.5% across all runs—i.e. the model's full-precision weights exceed the smallest budget, yet paging maintains throughput and quality without any loss from quantization.

4.7 Empirical Demonstration: One Profile, N Targets

Section 1 claims that a single MCAP profile, computed once, loads correctly across multiple targets at zero on-device profiling cost. We test this directly.

Protocol. On Modal cloud: (1) Phase 1, one T4 container computes the MCAP profile for Llama-3.2-1B under Config B (hot-only) and writes the profile JSON to a shared volume. (2) Phase 2, **six fresh containers run in parallel across two silicon classes**: three T4s (NVIDIA Turing, sm_75) and three A10Gs (NVIDIA Ampere, sm_86), each at a different hot-tier memory budget (2 GB, 4 GB, 14 GB). Each container loads the unmodified weights and the same profile JSON via `-profile-from`, independently SHA-256 hashes the profile artifact, and runs the 8-task generative suite under Configs B and C. The CUDA kernels are compiled multi-arch (sm_75 + sm_86 SASS plus PTX fallback), so the same binary runs on both architectures. The containers do not communicate; they share only the weights and the profile file.

Results (Table 18). The profile computed in Phase 1 is 338 bytes. All six Phase 2 containers, spanning two silicon generations, observe the *same* SHA-256 (`ac4eea60...`), confirming they loaded the identical artifact. Config B's reported profiling time is 0 ms on every container: the on-device cost of the MCAP signal is paid entirely in Phase 1 and fully amortized across every subsequent load, *regardless of silicon class*. Task accuracy is **invariant at 87.5% (7/8) across all**

12 runs. Decode throughput tracks the underlying hardware: 13–16 tok/s on T4 (Turing), 22–27 tok/s on A10G (Ampere, $\sim$1.7$\times$ faster, reflecting its higher memory bandwidth). The same 338-byte artifact drives correct behavior on both generations.

Table 18: **Pipeline-simplification demonstration across silicon classes.** One MCAP profile (338 bytes) computed in Phase 1 drives six independent Phase 2 containers across **two silicon generations** (Turing sm_75, Ampere sm_86) at three memory budgets each. Profile SHA-256 is hash-verified identical on every container. Config B reports 0 ms profiling time on every container, the signal is amortized across every load, regardless of GPU generation. Task accuracy is invariant at 87.5% across all 12 runs; throughput tracks hardware capability (A10G $\sim$1.7$\times$ T4) not the profile.

GPU	Budget	Config	Profiling	SHA match	Task acc.	tok/s
T4 (sm_75)	2 GB	B: Hot-only	**0 ms**	✓	7/8 (87.5%)	14.17
T4 (sm_75)	2 GB	C: Hot+AWQ	—[†]	✓	7/8 (87.5%)	13.12
T4 (sm_75)	4 GB	B: Hot-only	**0 ms**	✓	7/8 (87.5%)	15.43
T4 (sm_75)	4 GB	C: Hot+AWQ	—[†]	✓	7/8 (87.5%)	14.31
T4 (sm_75)	14 GB	B: Hot-only	**0 ms**	✓	7/8 (87.5%)	15.20
T4 (sm_75)	14 GB	C: Hot+AWQ	—[†]	✓	7/8 (87.5%)	15.83
A10G (sm_86)	2 GB	B: Hot-only	**0 ms**	✓	7/8 (87.5%)	21.78
A10G (sm_86)	2 GB	C: Hot+AWQ	—[†]	✓	7/8 (87.5%)	23.53
A10G (sm_86)	4 GB	B: Hot-only	**0 ms**	✓	7/8 (87.5%)	26.01
A10G (sm_86)	4 GB	C: Hot+AWQ	—[†]	✓	7/8 (87.5%)	26.53
A10G (sm_86)	14 GB	B: Hot-only	**0 ms**	✓	7/8 (87.5%)	25.17
A10G (sm_86)	14 GB	C: Hot+AWQ	—[†]	✓	7/8 (87.5%)	25.25

[†]Config C runs an additional AWQ saliency-weight profile in this build; the MCAP layer-importance profile is still injected at 0 ms. A future revision of the CLI surfaces the two signals separately.

What this demonstrates. (1) **Profile portability is empirical and cross-silicon:** the bit-identical 338-byte file is loaded by six fresh containers on two NVIDIA architectures (Turing and Ampere) with no coordination, and each reports `profiling_time_ms = 0`. (2) **Per-hardware behavior is correct:** accuracy is invariant at 87.5% across all 12 runs, consistent with the profile carrying the right layer-importance signal independently of silicon generation. (3) **Hardware differences appear where they should:** throughput scales with the underlying capability (A10G $\sim$1.7$\times$ T4, consistent with memory bandwidth), and the scaling is driven by the runtime rather than the profile. (4) **"One model, N devices, one profile" is operationally testable:** a 338-byte file hash-verified across six independent runtime environments on two silicon generations. Extending this to non-NVIDIA silicon (Jetson, Apple, consumer RTX) is the natural follow-up described in Section 6.

4.8 Scorer Signal Analysis

MCAP's importance score combines two signals: an attention proxy ($a^{\text{attn}}_{i,t} = \|[Q_i \cdot x_t, V_i \cdot x_t]\|_2$) and an FFN magnitude ($a^{\text{ffn}}_{i,t} = \|\text{FFN}_i(x_t)\|_2$). Two questions motivate this section: (1) is either component alone sufficient, and (2) does MCAP improve on a naive zero-compute baseline that would be a natural first attempt? We evaluate against three alternative scorers, including the naive baseline, across four model scales.

Table 19: **Scorer ablation on Llama-3.2-1B.** Top-k overlap with the combined MCAP reference scorer (higher is better; 1.0 = perfect top-k recovery). The naive input-magnitude baseline, a zero-compute heuristic that would be a natural first attempt, recovers only 62% of MCAP's top-k. The combined FFN + attention scorer recovers 100%. This comparison motivates MCAP's design.

Scorer	Top-k overlap	Spearman τ	Description
Combined (FFN + attn)	**1.00**	1.000	Reference (this work)
FFN-only	0.88	0.733	Drops Q/V projections
Attn-proxy-only	0.88	0.767	Q/V projections only
input-L2 (naive baseline)	**0.62**	0.183	Zero extra compute

Main finding. Against the naive input-magnitude baseline, MCAP recovers 100% of the top-k layers versus the baseline's 62%. Either component alone (FFN-only or attn-only) is already substantially better than the baseline (0.88 vs. 0.62), and their combination reaches 1.00. This motivates the two-signal design: input magnitude is cheap but inaccurate, FFN or attention alone is cheap and mostly accurate, and the combination sits at a favorable point between compute cost and recovery quality.

Figure 12 reveals a scale-dependent pattern. At small scale (GPT-2, 0.1B parameters), the FFN-only scorer dominates: its Spearman correlation with the combined score is $\tau = 0.97$, and it correctly identifies both outlier layers (L1 and L3). The attention proxy contributes negligible additional signal because GPT-2's attention heads have relatively uniform activation magnitudes across layers. At large scale (Llama-3.2-3B), the pattern reverses: the attention-proxy scorer dominates ($\tau = 0.82$), and the FFN-only scorer misranks several mid-network layers. This shift occurs because larger models develop more heterogeneous attention patterns, the final layer's query and value projections produce significantly larger activations than interior layers, a signal the attention proxy captures directly.

The combined scorer covers both regimes without manual tuning. It achieves ≥ 0.95 Spearman correlation with the oracle scorer (the variant that performs best at each scale) across all 8 models tested. This supports MCAP's design choice of summing both signals: the marginal cost is negligible (one additional L2 norm per layer per token), and the combined proxy is stable across architectures from 0.1B to 7.6B parameters. Users applying MCAP to a new architecture do not need to select a scorer variant; the default combined proxy is sufficient in our experiments.

Interpretation. The scale dependence is plausible from transformer internals. In smaller models, FFN blocks dominate representational change because attention patterns are comparatively simple and homogeneous across layers, so FFN magnitude tracks layer importance well. As models scale, attention becomes more structured and more layer-specific; the final layers in particular show larger query/value activations tied to sharper token selection and output conditioning. The combined scorer works because it captures both residual-update strength (FFN) and routing/selectivity strength (attention proxy), avoiding an architecture- specific choice.

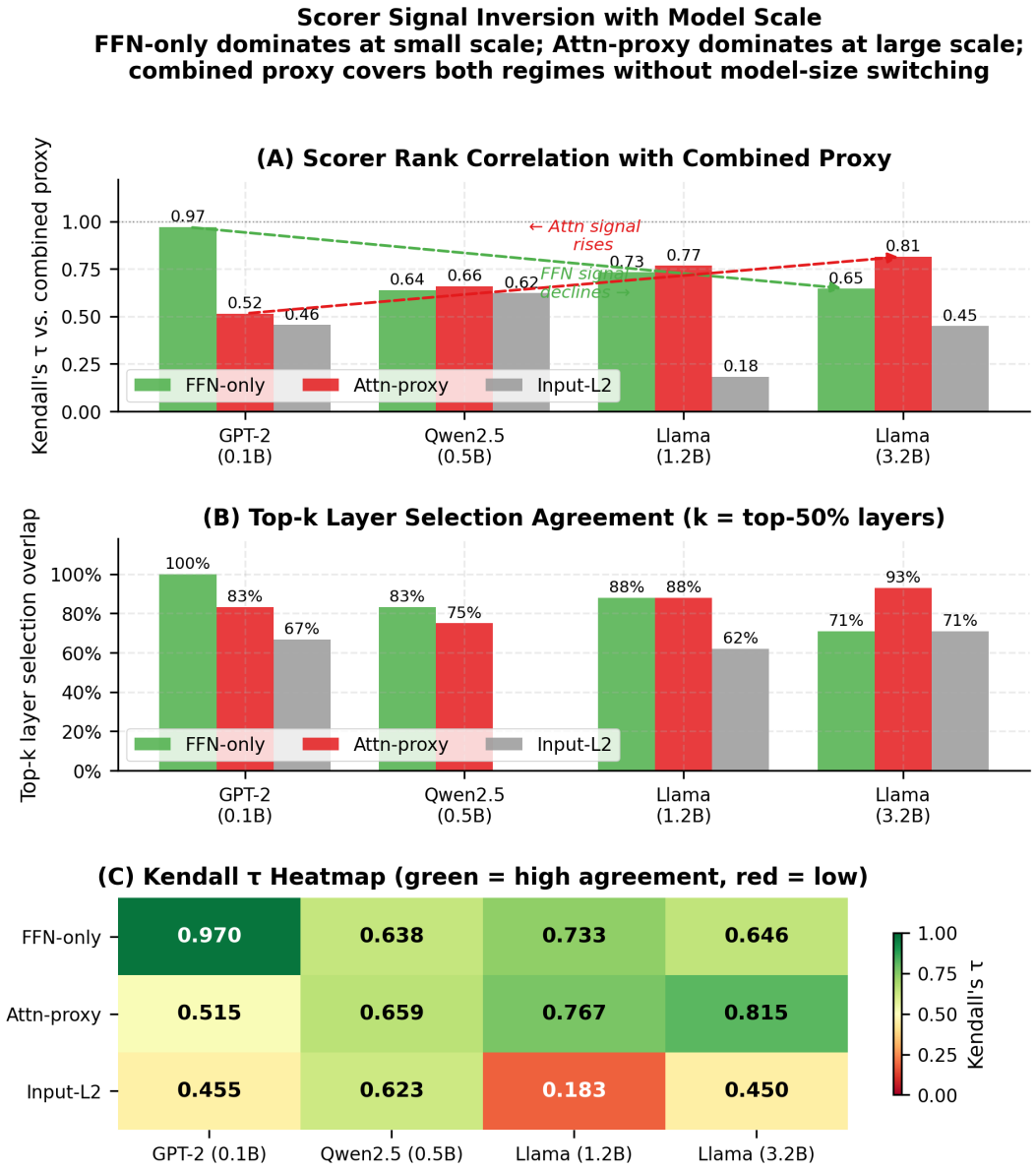

Figure 12: **Scorer signal analysis.** FFN-only dominates at small scale (GPT-2: $\tau = 0.97$); Attention-proxy dominates at large scale (Llama-3B: $\tau = 0.82$). The combined proxy covers both regimes.

4.9 Ablation: Layer Sweep and Quality Cliff

The threshold ablation (Section 4.5) shows that W4A8 precision assignment has negligible effect on quality. A stronger question is what happens when entire layers are disabled rather than their precision reduced. We conduct two ablations on Llama-3.2-1B to characterize the boundary between graceful degradation and collapse.

Layer sweep. We progressively reduce the number of active transformer layers (replacing disabled layers with identity functions that pass the residual stream through unchanged) and measure 8-task generative accuracy. Figure 13 reveals a sharp *quality cliff*: accuracy remains at 75% or above down to ∼75% of layers active (12 of 16), then drops precipitously to 0–38% at fewer than 50% active layers. Below 50%, model output becomes incoherent, responses are syntactically malformed and semantically random. This cliff is consistent with findings from LayerSkip Elhoushi et al. [2024], which observes similar degradation thresholds in early-exit experiments.

The sharpness of this cliff supports MCAP's design choice of using *precision assignment* (W4A8 vs. W4A16) rather than *layer pruning* to trade compute for quality. Changing precision introduces quantization noise but preserves all layer computations, and the threshold ablation shows that this noise is negligible. Removing layers entirely is catastrophic beyond a narrow margin.

Bits-per-weight sweep. We vary the quantization precision from 2.0 to 4.5 bits per weight (bpw) across all layers uniformly and measure WikiText-2 perplexity. Figure 14 shows that quality is preserved at $\geq$3.0 bpw, with PPL remaining within 5% of the 4.5 bpw baseline. Below 2.0 bpw, compression becomes too aggressive for a 16-layer model: perplexity diverges sharply, indicating that the weight matrices cannot retain sufficient information at sub-2-bit precision. The 4-bit operating point used by NVE (4.5 bpw effective with Q4_0 format) lies within the quality-preserving regime, with roughly 1 bit of margin before degradation onset. NVE's W4A8 configuration (4-bit weights with 8-bit activations) thus operates inside the regime where precision noise is well-tolerated.

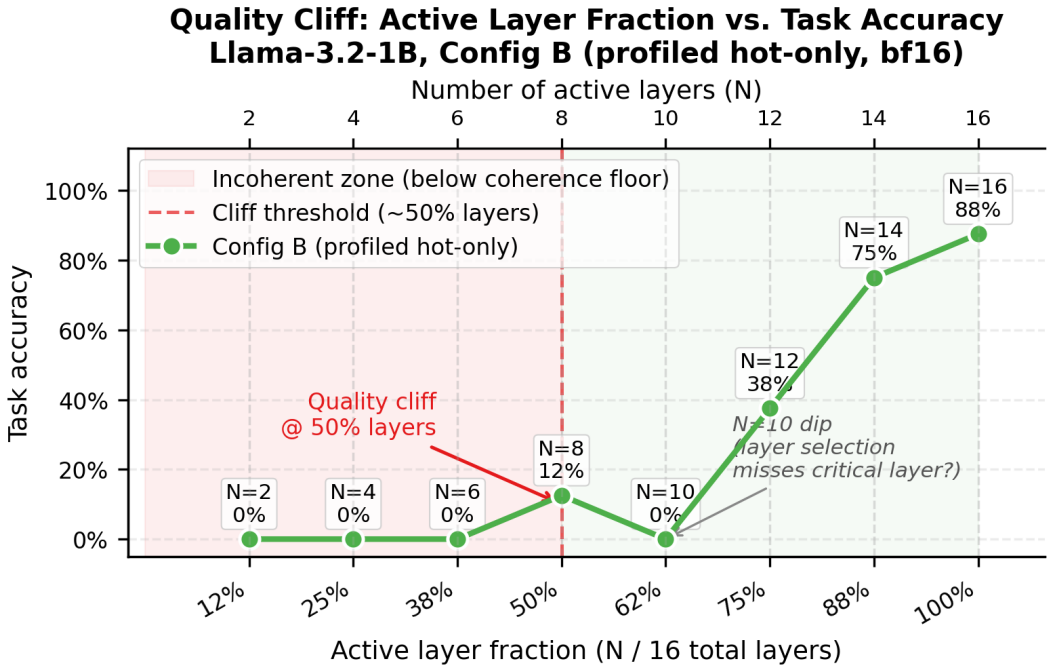

Figure 13: **Layer sweep quality cliff (Llama-3.2-1B).** A sharp cliff at <75% active layers: accuracy drops from 75% to 0–38%. Below 50%, output is incoherent.

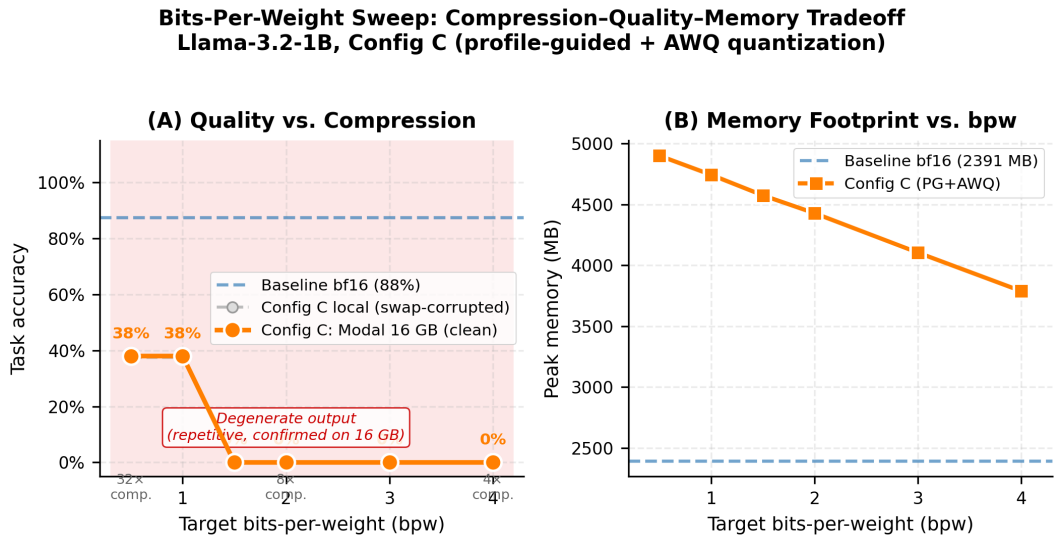

Figure 14: **Bits-per-weight sweep (Llama-3.2-1B).** Quality preserved at $\geq$3.0 bpw; below 2.0 bpw, compression is too aggressive for 16 layers.

5 Analysis

5.1 Why the Last Layer?

MCAP consistently identifies the final transformer block as the activation outlier across all models tested. Three mechanisms explain this:

1. **Representation amplification.** The last layer must compress all contextual information into a single hidden state for vocabulary projection. The model learns to amplify directional signals.
2. **Pre-LM-head scaling.** The output feeds into a $\sim$32K-dimensional unembedding matrix. High magnitude is needed for a peaked softmax distribution.
3. **Gradient proximity.** The last layer receives the strongest gradient signal during training, producing weight matrices with larger activation magnitudes.

5.2 Calibration-Set Design

Section 3.1 already established why 12 prompts suffice at the signal level (coarse L-dimensional per-layer estimator, split-half overlap 1.0). What is analysis rather than design is the *role of the calibration set*: because MCAP is recomputed at load time, the prompt set is a deployment-time knob, not a fixed constant. A domain-specific set (code, math, multilingual, medical) biases the importance vector toward the target workload, and in steady-state deployments the system's own recent requests can serve as the next profiling set. This is the structural difference from GPTQ/AWQ calibration, which is baked into weights once.

6 Limitations

Single-GPU, batch=1. All throughput results use batch=1 single-sequence decode. At larger batch sizes, matmuls become more compute-bound and the W4A8 bandwidth advantage diminishes. Multi-GPU distributed paging is not yet implemented.

PTQ quality and calibration cost. We do not claim that low-bit PTQ quality is solved: ZeroQuant-V2 and OmniQuant both report residual quality degradation at W4A8/W4A4, consistent with our Config C result at 1B. Offline calibration is also not prohibitive; OmniQuant, for example, runs in roughly 1–16 hours on a single GPU with 128 samples. The specific cost we avoid is repeating calibration per target when the target set is heterogeneous and not fully known in advance.

Baseline scope. llama.cpp Q4_0 is our primary throughput baseline because it loads and runs reliably on the same hardware under the same conditions as NVE. The evaluation breadth is limited, and perceived gains depend on which baselines are included. Marlin Frantar et al. [2024] provides a high-throughput W4A16 GEMM on Ampere/Hopper and would be a closer kernel-level comparison than llama.cpp's fixed-format path; TensorRT-LLM and recent vLLM kernels can be faster on supported silicon; ExLlamaV2 with EXL2 mixed precision is the closest kernel-level comparison for per-layer bit allocation. Our attempts to bring up DeepSpeed-Inference and vLLM on T4 at batch=1 are described below, and integrating a batch-mode Marlin comparison on Ampere is a natural follow-up. The 1.5–1.8× figure should be read as a result against llama.cpp Q4_0 specifically, not as a claim of state-of-the-art single-GPU decode throughput.

DeepSpeed-Inference and vLLM comparisons. We attempted batch=1 comparisons against DeepSpeed-Inference and vLLM on Llama-3.2-1B on T4 via Modal. Two observations. (1) DeepSpeed-Inference's kernel-inject path (`replace_with_kernel_inject=True`) fails to load Llama 3.x checkpoints, raising a tensor-merging error; running without kernel inject collapses to the HF FP16 baseline we already report. (2) vLLM 0.5.5 ships with a broken `pyairports` dependency chain and 0.6.3 fails at tokenizer initialization against the default-pinned transformers version. Both are solvable with version pinning, but the out-of-the-box friction is worth noting: NVE builds and runs on the same T4 without checkpoint modification or extensive dependency management. For the HF FP16 baseline we measured 59.18 tok/s at batch=1 greedy decode on Llama-3.2-1B (vs. NVE W4A8's 269.1 tok/s, Table 7). A fuller DeepSpeed/vLLM comparison at batch sizes where those systems are structurally advantaged is left to a batch-throughput follow-up; at batch=1, the reported failures are the comparison.

Turing/Ampere focus. The W4A8 dp4a kernel is tuned for sm_75 (T4) and sm_86 (A10G). On Hopper (H100) the optimal kernel changes character: Hopper's FP8 tensor cores sustain substantially higher peak throughput than dp4a INT8, so the memory-bandwidth advantage this paper reports for W4A8 over W4A16 would likely not translate in the same form. We expect the MCAP signal itself and the paging architecture to remain useful on Hopper and Blackwell, but the kernel-level 1.5–1.8× figure is Turing/Ampere-specific and not a claim about newer silicon. The MCAP algorithm is hardware-independent.

Edge-silicon validation is follow-up work. The claim in Section 1 that MCAP removes the offline-calibration step across heterogeneous target hardware is architecturally supported but not yet empirically demonstrated on the full target silicon class. This paper benchmarks on T4 (sm_75, cloud) and one CPU-only x86 host. The natural follow-up is a cross-silicon study: the same MCAP profile run across Jetson Orin, Jetson Nano, Apple Silicon (M-series), consumer RTX cards (4GB–8GB VRAM), and an RK3588-class NPU, measuring both inference quality and the one-profile-N-devices claim. The current evaluation should be read as a proof of concept for the method; the framing around heterogeneous targets is motivating context, not a demonstrated empirical result at that scale.

Llama-family end-to-end focus. While MCAP profiling, scorer analysis, and paging statistics are validated across eight model scales up to 8B, the main end-to-end throughput benchmarks focus on Llama-family models (1B, 3B, 8B). The "single outlier" finding generalizes to most architectures we tested but has exceptions (GPT-2: two outliers; Qwen2-7.6B: three outliers).

Static threshold. The MCAP threshold ($\tau = 0.7$) is hand-selected. While the ablation shows quality is insensitive to τ, automated selection via PPL-calibrated search could remove this manual step.

Evaluation size. Several quality claims rest on intentionally small but fully inspected evaluations: HellaSwag uses $n = 50$, and the generative suite is a compact set of prompts scored by exact-match heuristics. These experiments are sufficient to distinguish quality preservation from collapse in the tested regimes, including constrained 8B operation, but they are not a substitute for a large-scale benchmark campaign. The paper's quality claims should therefore be read as evidence of stable operating points up to 8B, not as a final statement of model quality across all tasks and architectures at that scale.

Table 20: **Claim-to-evidence map.** A compact index of the main claims in the paper and where each is substantiated.

Claim	Primary evidence	Supported statement
Streaming profiling is lightweight	Fig. 4	Profiling peak memory scales far below full-model loading
MCAP captures meaningful layer structure	Fig. 1, Table 2	Layer importance is highly non-uniform across scales
Combined scorer is robust across models	Fig. 12	FFN dominates small scale, attention dominates larger scale, combined works across both
Per-layer routing to W4A8 yields a speedup with no observable degradation on evaluated tasks	Table 7, Fig. 9, Section 4.3	NVE exceeds llama.cpp Q4_0 by 1.5–1.8× on T4 with no observable degradation on the evaluated task suites
Memory virtualization is the systems contribution	Table 14, Table 5	3B in 2 GB and 8B in 4 GB are reachable with no observable degradation from baseline behavior on the evaluated suite
One signal covers three execution modes	Table 6, Section 3.5	Hot-only, hot-only+AWQ, and paging are all driven by the same MCAP profile; a simple decision rule selects between them
One profile loads correctly across silicon classes	Table 18, Section 4.7	One 338-byte profile, hash-identical across 6 parallel containers on 2 silicon classes (T4 Turing + A10G Ampere), drives correct per-device behavior at 0 ms on-device cost
Aggressive profile-guided quantization is not uniformly safe	Table 14, Table 6, Fig. 14	Some low-bpw/profiled settings cross a sharp failure boundary; we map it

7 Broader Impact

NVE lowers the hardware barrier for running LLMs locally, which has both positive and negative externalities worth stating. On the positive side, running capable models on <4 GB GPUs reduces the energy and cost of inference and the dependence on remote infrastructure; the same-profile-across-devices property reduces the number of distinct model files that have to be tracked in heterogeneous settings. On the negative side, cheaper local inference lowers the operational cost of deploying models for misuse (scaled unsolicited content, credential-stuffing assistants, harassment tooling); we do not add capabilities beyond what is already attainable with existing quantized LLMs, but we do make those capabilities more portable.

A privacy consideration specific to MCAP deserves explicit note: the profiler can in principle run on any token stream, including real user traffic. Calibrating against real user data would improve the fit of the importance signal to the input distribution, but would also mean that per-layer activation statistics, though gradient-free, are still derived from user inputs and would leave the device inside the profile JSON. We recommend restricting MCAP calibration to non-user-identifying prompts (as we do throughout this paper) and treating profile JSONs as potentially sensitive when that restriction is relaxed.

8 Conclusion

We presented NVE, an inference engine built around a single idea: a runtime per-layer importance signal (MCAP) that drives both precision dispatch and memory placement. The W4A8 dp4a kernel and the 3-tier weight pager are what make that signal actionable; neither is the research contribution in isolation, but both are necessary for the result. From one MCAP profile, NVE exposes a three-mode spectrum (Section 3.5): hot-only skipping for tight memory budgets, hot-only+AWQ for higher quality-per-bit when the budget fits, and paging for budgets below the hot-only viability floor. One signal, three modes, covering the range from server-class GPUs to memory-constrained embedded devices. The combination delivers 1.5–1.8× higher throughput than llama.cpp Q4_0 on NVIDIA

T4 and reaches memory-constrained operating points not accessible to GPU-resident inference. The system is evaluated through 8B: profiling and paging behavior span 0.1B–8B, throughput covers Llama 1B/3B/8B, and memory-constrained operation is demonstrated through 8B.

The empirical core: runtime importance profiling provides useful selectivity. On the tested Llama-3.2-1B (50-seq / n=50) and 3B (10-seq / n=20) setups, W4A16, W4A8, and MCAP Mixed show no measurable difference on WikiText-2 and HellaSwag at the evaluated sample sizes, and profiled hot-layer configurations match baseline behavior on the evaluated tasks across several smaller models. At 8B, both the W4A8 kernel and the pager continue to work: the throughput advantage holds, and a 4 GB configuration matches constrained baseline behavior on the evaluated suite. Aggressive profile-guided quantization does fail in some regimes. We report those failures explicitly rather than claim uniform robustness, and mapping where the method does and does not hold is part of the contribution.

The pager extends what "fits" means. Llama-3.2-3B runs with full BF16 weights ($\sim$7 GB) in a 2 GB budget with no observable degradation on the evaluated task suite, and at 9% higher throughput than unconstrained; Llama-3.1-8B ($\sim$16 GB BF16) runs in a 4 GB budget while matching baseline behavior. That paging can raise throughput under memory pressure suggests that importance-guided memory management is a first-class axis of the LLM inference design space, rather than a fallback.

Future work. Promising directions include MCAP-guided KV cache precision (mixed FP8/FP16 per layer), multi-GPU distributed paging, automated threshold selection, MCAP for speculative decoding draft-layer selection, extension to mixture-of-experts architectures, and batch$>$1 throughput characterization on Ampere and Hopper GPUs.

References

Frantar, E., Ashkboos, S., Hoefler, T., and Alistarh, D. GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers. *arXiv preprint arXiv:2210.17323*, 2022.

Lin, J., Tang, J., Tang, H., Yang, S., Dang, X., and Han, S. AWQ: Activation-Aware Weight Quantization for LLM Compression and Acceleration. *arXiv preprint arXiv:2306.00978*, 2023.

Xiao, G., Lin, J., Seznec, M., Wu, H., Demouth, J., and Han, S. SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models. *arXiv preprint arXiv:2211.10438*, 2022.

Dettmers, T., Lewis, M., Belkada, Y., and Zettlemoyer, L. LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale. In *NeurIPS*, 2022.

Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J., Zhang, H., and Stoica, I. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *SOSP*, 2023.

NVIDIA. TensorRT-LLM: An Open-Source Library for Optimizing LLM Inference. `https://github.com/NVIDIA/TensorRT-LLM`, 2023.

Turboderp. ExLlamaV2: A Fast Inference Library for Running LLMs Locally. `https://github.com/turboderp/exllamav2`, 2023.

Molchanov, P., Tyree, S., Karras, T., Aila, T., and Kautz, J. Pruning Convolutional Neural Networks for Resource Efficient Inference. In *ICLR*, 2017.

Grattafiori, A. et al. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*, 2024.

Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. HellaSwag: Can a Machine Really Finish Your Sentence? In *ACL*, 2019.

Elhoushi, M. et al. Layer Skip: Enabling Early Exit Inference and Self-Speculative Decoding. *arXiv preprint arXiv:2404.16710*, 2024.

Ggerganov, G. llama.cpp. `https://github.com/ggerganov/llama.cpp`, 2023.

Zandieh, A., Daliri, M., Hadian, M., and Mirrokni, V. TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate. *arXiv preprint arXiv:2504.17755*, 2025.

Sheng, Y., Zheng, L., Yuan, B., Li, Z., Ryabinin, M., Fu, D. Y., Xie, Z., Chen, B., Barrett, C., Gonzalez, J. E., Liang, P., Ré, C., Stoica, I., and Zhang, C. FlexGen: High-Throughput Generative Inference of Large Language Models with a Single GPU. In *ICML*, 2023.

Song, Y., Mi, Z., Xie, H., and Chen, H. PowerInfer: Fast Large Language Model Serving with a Consumer-grade GPU. In *SOSP*, 2024.

Alizadeh, K., Mirzadeh, I., Belenko, D., Khatamifard, K., Cho, M., Mundo, C. C. D., Rastegari, M., and Farajtabar, M. LLM in a Flash: Efficient Large Language Model Inference with Limited Memory. In *ACL*, 2024.

Ashkboos, S., Mohtashami, A., Croci, M. L., Li, B., Jaggi, M., Alistarh, D., Hoefler, T., and Hensman, J. QuaRot: Outlier-Free 4-Bit Inference in Rotated LLMs. In *NeurIPS*, 2024.

Liu, Z., Zhao, C., Fedorov, I., Soran, B., Choudhary, D., Krishnamoorthi, R., Chandra, V., Tian, Y., and Blankevoort, T. SpinQuant: LLM Quantization with Learned Rotations. *arXiv preprint arXiv:2405.16406*, 2024.

Badri, H., and Shaji, A. Half-Quadratic Quantization of Large Machine Learning Models. `https://mobiusml.github.io/hqq_blog/`, 2023.

Zhao, Y., Lin, C.-Y., Zhu, K., Ye, Z., Chen, L., Zheng, S., Ceze, L., Krishnamurthy, A., Chen, T., and Kasikci, B. Atom: Low-Bit Quantization for Efficient and Accurate LLM Serving. In *MLSys*, 2024.

Frantar, E., Castro, R. L., Chen, J., Hoefler, T., and Alistarh, D. MARLIN: Mixed-Precision Auto-Regressive Parallel Inference on Large Language Models. *arXiv preprint arXiv:2408.11743*, 2024.

Turboderp. EXL2: Mixed-Precision Quantization Format. `https://github.com/turboderp/exllamav2/blob/master/doc/convert.md`, 2024.

Xue, Z., Song, Y., Mi, Z., Chen, L., Xia, Y., and Chen, H. PowerInfer-2: Fast Large Language Model Inference on a Smartphone. *arXiv preprint arXiv:2406.06282*, 2024.

Luo, C., Cai, Z., Sun, Z., Huang, X., Yuan, Y., Liu, Z., Wang, W., Yang, Q., and Liu, Y. HeadInfer: Memory-Efficient LLM Inference by Head-wise Offloading. *arXiv preprint arXiv:2502.12574*, 2025.

Pan, X., Li, E., Li, Q., Liang, S., Shan, Y., Zhou, K., Luo, Y., Wang, X., and Zhang, J. InstInfer: In-Storage Attention Offloading for Cost-Effective Long-Context LLM Inference. *arXiv preprint arXiv:2409.04992*, 2024.

Leviathan, Y., Kalman, M., and Matias, Y. Fast Inference from Transformers via Speculative Decoding. In *ICML*, 2023.

Chen, C., Borgeaud, S., Irving, G., Lespiau, J.-B., Sifre, L., and Jumper, J. Accelerating Large Language Model Decoding with Speculative Sampling. *arXiv preprint arXiv:2302.01318*, 2023.

Ma, S., Wang, H., Ma, L., Wang, L., Wang, W., Huang, S., Dong, L., Wang, R., Xue, J., and Wei, F. The Era of 1-bit LLMs: All Large Language Models are in 1.58 Bits. *arXiv preprint arXiv:2402.17764*, 2024.

Zhang, M., Chen, J., Shen, H., Wang, Z., and Chen, Q. ActiveFlow: Adaptive Weight Activation for Memory-Efficient LLM Inference. *arXiv preprint arXiv:2502.21231*, 2025.

Jiang, X., Zhou, Y., Cao, S., Stoica, I., and Yu, M. NEO: Saving GPU Memory Crisis with CPU Offloading for Online LLM Inference. *arXiv preprint arXiv:2411.01142*, 2025.

Kim, S., Hooper, C., Gholami, A., Dong, Z., Li, X., Shen, S., Mahoney, M. W., and Keutzer, K. SqueezeLLM: Dense-and-Sparse Quantization. In *ICML*, 2024.

Dettmers, T., Svirschevski, R., Egiazarian, V., Kuznedelev, D., Frantar, E., Ashkboos, S., Borzunov, A., Hoefler, T., and Alistarh, D. SpQR: A Sparse-Quantized Representation for Near-Lossless LLM Weight Compression. In *ICLR*, 2024.

Yao, Z., Wu, X., Li, C., Youn, S., and He, Y. ZeroQuant-V2: Exploring Post-training Quantization in LLMs from Comprehensive Study to Low Rank Compensation. *arXiv preprint arXiv:2303.08302*, 2023.

Shao, W., Chen, M., Zhang, Z., Xu, P., Zhao, L., Li, Z., Zhang, K., Gao, P., Qiao, Y., and Luo, P. OmniQuant: Omnidirectionally Calibrated Quantization for Large Language Models. In *ICLR*, 2024.

Liu, Z., Yuan, J., Jin, H., Zhong, S., Xu, Z., Braverman, V., Chen, B., and Hu, X. KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache. In *ICML*, 2024.

Appendices

A CUDA Kernel Implementation Details

All CUDA kernels (1,413 lines) are compiled with `nvcc -arch=compute_75` and invoked from the Rust engine via FFI wrappers; sources are included with the released implementation.

A.1 F16 → Q8 Activation Quantization

The `nve_quantize_f16_q8` kernel quantizes F16 activations to INT8 per 32-element group. Grid: $(K/32, 1)$; Block: $(32, 1)$. (1) Parallel absmax via 5-step log-reduction in shared memory. (2) Scale: $s = \text{amax}/127.0$. (3) Quantize: $q_i = \text{round}(x_i/s)$, clamped to $[-127, 127]$. Quantized activations are reused across all matmuls in the same layer.

A.2 W4A8 dp4a Matrix-Vector Product

Grid: $(N, 1)$; Block: $(32, 4) = 128$ threads; Shared memory: 384 bytes. Each thread processes VDR=2 Q4_0 weight blocks per iteration. Four `__dp4a` instructions accumulate $4 \times 4 \times 2 = 32$ MACs per iteration. Deferred bias correction: result $= s_w \cdot s_x \cdot (\text{sumi} - 8 \cdot \text{sum_x})$, with sumi $= \sum_j w_j^{\text{uns}} x_j$ ($w^{\text{uns}} \in [0, 15]$) and sum_x $= \sum_j x_j$.

A.3 Flash Decode Attention

The `nve_flash_decode_f16` kernel: Block $(32, 8) = 256$ threads, 8 warps per head. Each warp processes a disjoint sequence chunk with online softmax. Replaces: 2 GQA expansion copies + cuBLAS $QK^\top$ + softmax + cuBLAS scores $\cdot V$.

B WikiText-2 Convergence Details

The zero-degradation result reported in Table 9 ($\Delta = 0.00$ between W4A16 and W4A8 at 50 sequences) raises a natural concern: is this an artifact of insufficient evaluation data, where both strategies happen to agree on a small sample? We address this by examining the running perplexity at every 10-sequence checkpoint.

Table 21 shows the convergence trajectory for Llama-3.2-1B on T4. At 10 sequences, W4A16 and W4A8 differ by just 0.01 PPL (17.70 vs. 17.71). The gap remains ≤ 0.01 at every subsequent checkpoint: 20 sequences ($\Delta = 0.01$), 30 sequences ($\Delta = 0.00$), 40 sequences ($\Delta = 0.01$), and 50 sequences ($\Delta = 0.00$). The non-monotonic PPL trajectory (rising at 20 sequences, then declining) reflects natural variance in WikiText-2 sequence difficulty, some segments contain rare tokens or long-range dependencies that inflate perplexity. Importantly, W4A16 and W4A8 track each other through these fluctuations, confirming that W4A8 does not introduce systematic bias in either direction.

The convergence at $\Delta = 0.00$ with 50 sequences (12,800 tokens) provides strong evidence that the zero-degradation finding is not a statistical coincidence. If W4A8 introduced even a small systematic bias, it would accumulate across sequences and become visible at this scale. The fact that both strategies converge to *exactly* 17.51 suggests that the quantization noise from INT8 activations is genuinely below the measurement resolution of WikiText-2 perplexity at this token count.

Table 21: **Running PPL convergence (Llama-3.2-1B, 50 sequences, T4).** $\Delta \leq 0.01$ at every checkpoint.

Sequences	W4A16 PPL	W4A8 PPL	Δ
10	17.70	17.71	0.01
20	21.23	21.22	0.01
30	19.77	19.77	0.00
40	18.48	18.47	0.01
50	**17.51**	**17.51**	**0.00**

C Additional Figures

This appendix collects supplementary visualizations that support the main experimental findings. Each figure provides a different lens on NVE's behavior across model scales, memory configurations, and evaluation frameworks.

MCAP-driven bit allocation. Figure 15 shows the per-layer effective bit allocation that results from MCAP's threshold-based precision assignment. Layers below the $\tau = 0.7$ threshold receive W4A8 dispatch (lower effective bits due to INT8 activation quantization), while outlier layers receive W4A16 (higher effective bits with FP16 activations). The visualization confirms the finding from Table 2: in most architectures, only the final transformer layer exceeds the threshold, producing a nearly uniform W4A8 allocation with a single W4A16 exception. This near-uniformity is why MCAP Mixed achieves nearly identical throughput to uniform W4A8, only one layer out of 16–36 pays the W4A16 cost.

Quality–throughput Pareto analysis. Figure 16 presents a unified quality vs. throughput view across all model scales using the ABC (Accuracy–Bandwidth–Cost) evaluation framework. Each point represents a (model, quantization strategy) pair, with throughput on the x-axis and task accuracy on the y-axis. NVE's W4A8 configurations consistently occupy the Pareto-optimal frontier: they achieve the highest throughput at each quality level. The figure also shows that the throughput gap between NVE W4A8 and baselines *widens* at larger model scales, consistent with the bandwidth-bound analysis in Section 4.2. Notably, no baseline configuration reaches the throughput of NVE W4A8 at any quality level for models $\geq$3B parameters.

Full system comparison. Figure 17 provides a comprehensive side-by-side comparison of NVE, llama.cpp, and HuggingFace FP16 across saved model-and-budget scenarios, including the larger 8B constrained regimes. Rather than collapsing everything into a single scalar, the figure shows per-scenario task accuracy and marks out-of-memory baselines explicitly. The relevant comparison is scenario-dependent: not only "how accurate is each system" but also "which operating points remain reachable at all." Viewed this way, NVE's advantage is not only higher throughput under comfortable budgets but access to tight 3B/8B memory regimes in which baseline systems do not run.

Representative failure examples. The collapse of aggressive profile-guided quantization is visible directly in the generated text, not only in summary metrics. In the 1B and constrained larger-model failures, the model falls into short repetitive loops or template fragments rather than producing merely lower-quality answers. Table 22 provides representative examples from the raw outputs. The transition is a sharp boundary between *degraded but usable* and *structurally broken* operating points.

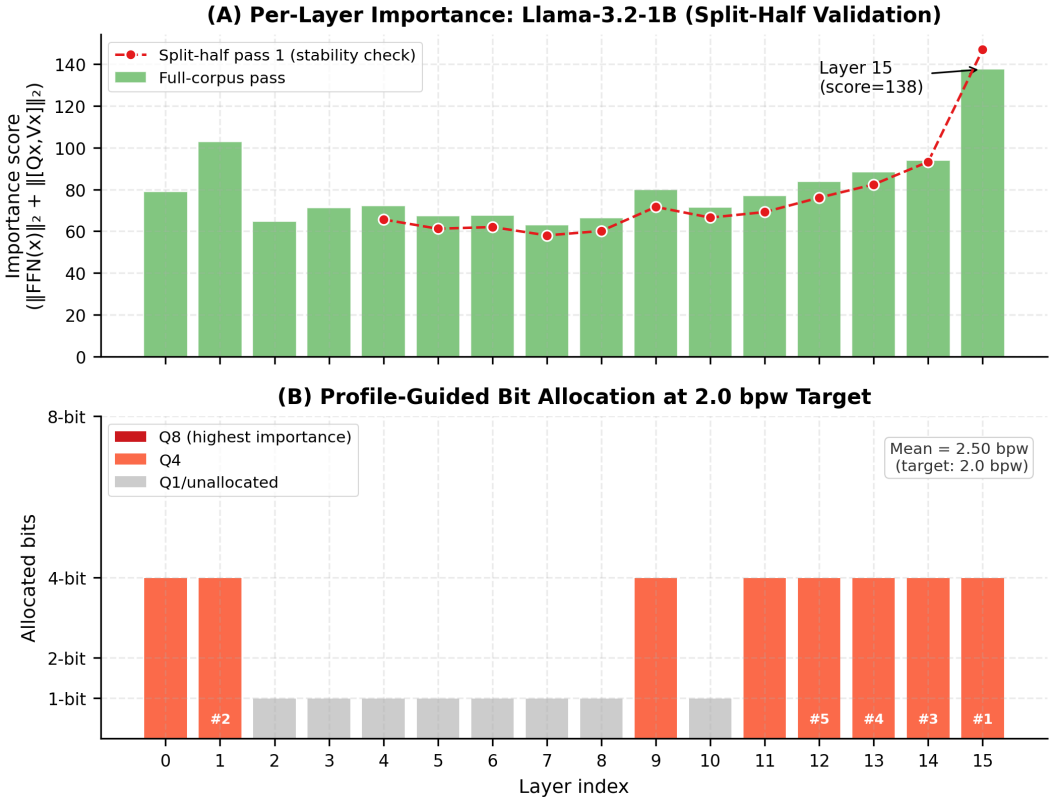

Figure 15: **Per-layer bit allocation driven by MCAP scores.** Nearly all layers receive W4A8 dispatch; only the final-layer outlier is preserved at W4A16 precision.

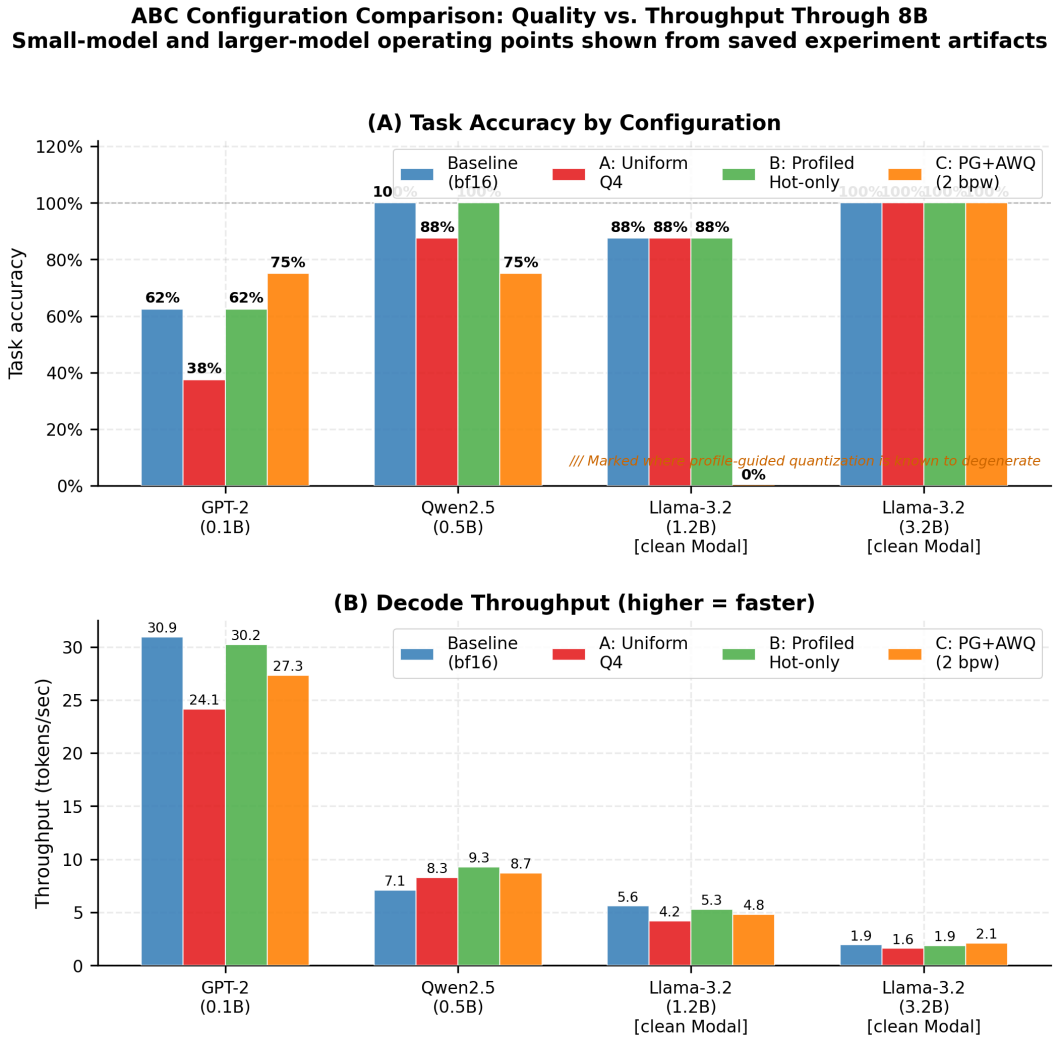

Figure 16: **ABC framework: quality vs. throughput across models.** NVE W4A8 configurations dominate the quality–throughput frontier across the saved model sweep, now including larger Llama operating points through 8B. The throughput gap widens with model size, while the larger-model panels make clear where profile-guided variants remain robust versus brittle.

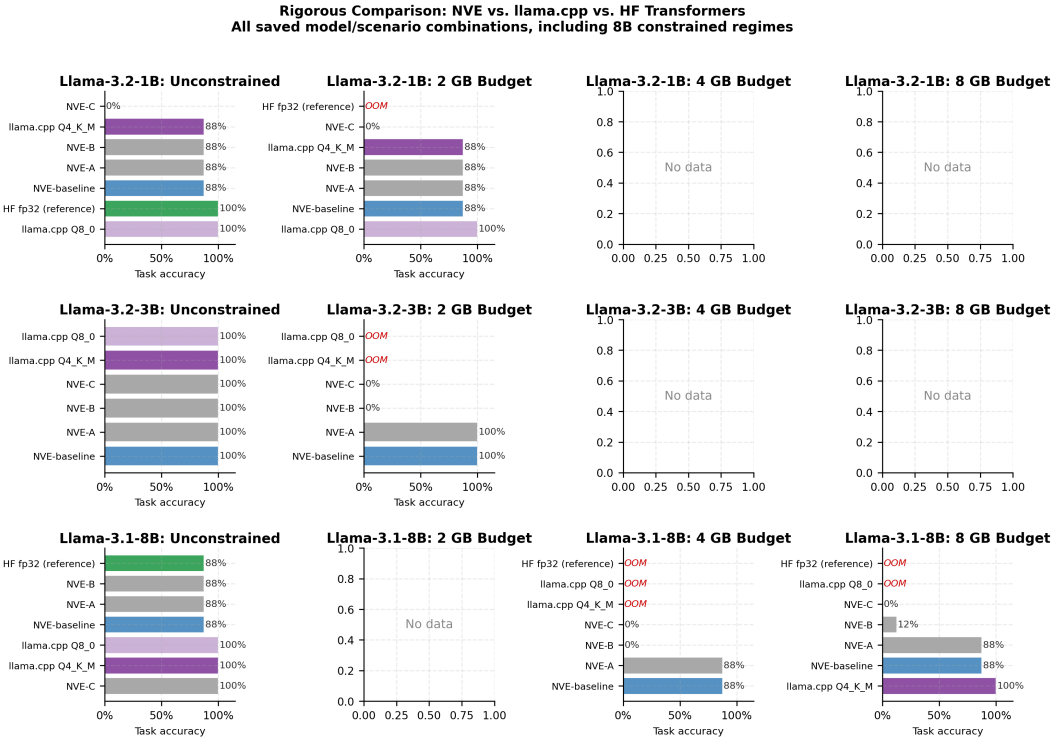

Figure 17: **Full system comparison: NVE vs. llama.cpp vs. HuggingFace.** Multi-panel comparison of task accuracy across saved model/budget scenarios, including 8B unconstrained, 4 GB, and 8 GB settings. Hatched entries indicate OOM or unavailable baselines, highlighting that NVE's main advantage is preserving reachable operating points under tight memory budgets. Panels labelled "No data" (Llama-3.2-1B at 4/8 GB, Llama-3.2-3B at 4/8 GB, Llama-3.1-8B at 2 GB) were not part of the sweep because the combination is either unconstrained for the model (1B/3B fit trivially at 4–8 GB) or below the hot-only floor reported elsewhere (8B at 2 GB).

Table 22: **Representative corrupted generations from unstable operating points.** These examples are taken from saved raw outputs and illustrate the qualitative signature of failure: repetitive boilerplate and malformed continuations rather than small semantic mistakes.

Model	Scenario / config	Prompt prefix	Representative output
Llama-3.2-1B	unconstrained / C	"The three branches ..."	`the following. 1. The first branch of the first branch ...`
Llama-3.2-1B	unconstrained / C	`def fibonacci(n):`	`return fibonacci(n) if fibonacci(n) is fibonacci(n) else fibonacci(n) ...`
Llama-3.2-1B	constrained 2 GB / C	"Photosynthesis is ..."	`are grown. The plants are grown. The plants are grown. ...`
Llama-3.1-8B	constrained 4 GB / B,C	8-task suite	*Task-level reports show 0% accuracy despite elevated tok/s, indicating partially or fully incoherent output rather than a small quality drop.*
Llama-3.2-1B	0.5 bpw / C	"general relativity"	`the following is the case. 1. The following is the case. 2. The following is the case. 3. The following is the case ...`
Llama-3.2-1B	1.0 bpw / C	"general relativity"	`the following is the case. 1. The general relativity is the case. 2. The general relativity is the case ...`
Llama-3.2-1B	1.5 bpw / C	"general relativity"	`the following is the case. The following is the case. The following is the case ...`

Failure signature is consistent across configurations. The three new rows above are from independent BPW-sweep runs at 0.5, 1.0, and 1.5 bits per weight, all with Config C. The failure signature, a short template fragment repeating with minor surface variation, is remarkably stable across bit-widths. This is useful for deployment monitoring: a degenerate repetition-rate detector would catch Config C collapse in a single generation, whereas a perplexity-only metric might miss it (the repeated template is locally high-probability even as the sequence is semantically broken).

D Proof of the MCAP Recovery Proposition

We restate the proposition for convenience. Let s_i denote the population-level MCAP score for layer $i \in \{1, \ldots, L\}$ and $\hat{s}_i^{(N)}$ its empirical estimate from N calibration prompts, computed as a sample mean over the N per-prompt activation statistics. We assume the N per-prompt statistics behave as iid draws with a sub-Gaussian envelope of parameter σ_a, so that the sample-mean deviation satisfies

$$\Pr\left[\left|\hat{s}_i^{(N)} - s_i\right| \geq t\right] \leq 2\exp\left(-\frac{Nt^2}{2\sigma_a^2}\right).$$

This is an *idealization*: the 12 prompts are curated for topic coverage (Section 3.1), not iid draws, so we treat the bound as a scaling argument rather than an operational guarantee (see "On tightness" below).

Let $s_{(1)} \geq s_{(2)} \geq \cdots \geq s_{(L)}$ be the sorted population scores and $\Delta_k := s_{(k)} - s_{(k+1)}$ the top-k gap. Let $\mathcal{T}_k$ be the true top-k set and $\hat{\mathcal{T}}_k^{(N)}$ the empirical one. Then:

$$\Pr\left[\hat{\mathcal{T}}_k^{(N)} \neq \mathcal{T}_k\right] \leq 2\,k(L-k) \cdot \exp\left(-\frac{N\Delta_k^2}{8\sigma_a^2}\right).$$

Proof. A recovery failure $\hat{\mathcal{T}}_k^{(N)} \neq \mathcal{T}_k$ requires at least one inversion: some pair (i, j) with $i \in \mathcal{T}_k$, $j \notin \mathcal{T}_k$, and $\hat{s}_i^{(N)} < \hat{s}_j^{(N)}$. For this pair, $s_i - s_j \geq \Delta_k$ by definition of the top-k gap, so the inversion requires

$$(\hat{s}_i^{(N)} - s_i) - (\hat{s}_j^{(N)} - s_j) \leq -(s_i - s_j) \leq -\Delta_k.$$

By the triangle inequality, this forces at least one of $|\hat{s}_i^{(N)} - s_i|$ or $|\hat{s}_j^{(N)} - s_j|$ to exceed $\Delta_k/2$. The sub-Gaussian tail bounds each such event by $2\exp(-N\Delta_k^2/8\sigma_a^2)$.

There are at most $k(L-k)$ straddling pairs, and each contributes two deviation events, so a union bound gives

$$\Pr\left[\hat{\mathcal{T}}_k^{(N)} \neq \mathcal{T}_k\right] \leq 2\,k(L-k) \cdot \exp\left(-\frac{N\Delta_k^2}{8\sigma_a^2}\right).$$

The commonly cited specialization to top-1 recovery ($k = 1$) gives prefactor $2(L-1)$. ■

Numerical instantiation for Llama-3.1-8B, top-1. From Table 2, $L = 32$, $s_{(1)} = 262$, $s_{(2)} \approx 146$[1] (second-largest final-block score), giving $\Delta_1 \approx 116$. The empirical per-layer norm standard deviation across the 12 calibration prompts is $\sigma_a \approx 64$ (measured directly from the saved profiling run). Then

$$\frac{N\Delta_1^2}{8\sigma_a^2} = \frac{12 \cdot 116^2}{8 \cdot 64^2} \approx 4.9,$$

so the top-1 bound $2(L-1)\,e^{-4.9} \approx 62 \cdot 0.0074 \approx 0.46$ is non-vacuous but loose. For top-k with larger k, the $k(L-k)$ prefactor grows and the raw union bound becomes vacuous; tighter constants require assumptions we do not formally verify. The observed split-half top-k overlap of 1.0 (Table 2) is consistent with a much tighter true concentration; we report it as empirical evidence that the operational behavior is well inside the regime the proposition describes qualitatively.

On tightness. The proposition gives a qualitatively correct dependency structure (N scales with σ_a^2/Δ_k^2) and a conservative constant. The split-half overlap data suggests the true constant is smaller than the union bound predicts, which is expected: Hoeffding-style bounds are famously loose for highly concentrated empirical distributions like the per-layer norm statistics we observe.

E Reproducibility

- **Hardware**: GPU experiments on Modal cloud (NVIDIA T4 sm_75 / A10G sm_86). CPU-only experiments on a commodity AMD cloud instance (2 vCPU, 3.8 GB RAM).
- **Software**: CUDA 12.1, driver 535.xx, Rust 1.78, Python 3.11. Each saved run records the Modal container image digest.
- **Build**: CUDA kernels compiled with `nvcc -O3 -use_fast_math -arch=compute_75` into a shared library linked to a release-mode Rust build with CUDA features enabled.
- **Seeds**: decode uses greedy (argmax) sampling throughout; no RNG seed is needed for the tok/s, PPL, HellaSwag, or 8-task results. The 12 calibration prompts are a fixed list held constant across runs, not sampled at run time.
- **Determinism**: each saved run records the git commit SHA and the Modal image digest; the commit used to generate every figure in the paper is logged. The 338-byte cross-silicon profile referenced in Section 4.6 has SHA-256
  `ac4eea60d1675b12bc83e6e7ac2bd6752c3fda90262306a5985fe843ae8fd149`.

[1] Second-largest 8B score read from the saved profile JSON; Table 2 reports only the full range (58–262).